Financial Intermediation and Asset Price Dynamics in the Post-COVID Monetary Expansion

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List of Abbreviations

DFR	Deposit facility rate
EA	Euro Area
ECB	European Central Bank
GFC	Global Financial Crisis
IRF	Impulse response function
LP	Local projection
NFC	Non-financial corporation
SVAR	Structural vector autoregression
VAR	Vector autoregression

List of Symbols

$\alpha,\ \beta$	Generic scalar coefficients
$h,\ \mathcal{H}$	Response horizon and the horizon set $\{0, 1, 3, 6, 12, 24\}$
z_e^f	High-frequency surprise for factor family f at ECB event e
$Z_m^f,\ \mathcal{E}(m)$	Monthly aggregate surprise for factor f, and the ECB events in month m
Shock_t	Easing-signed monthly policy-news shock, $-Z_m^{target}$ for month m
$\alpha_h,\ \beta_h$	Local-projection intercept and shock-response coefficient at horizon h
$X_t,\ \Gamma_h$	Predetermined controls and control-coefficient vector
ε_{t+h}	Local-projection residual at horizon h
$\widehat{\text{IRF}}_h$	Normalized impulse response at horizon h
$\widehat{\sigma}_{\text{Shock}}$	Estimated standard deviation of the shock
$\text{CIRF}_i(H)$	Cumulative impulse response of variable i through horizon H
$\lambda,\ \theta,\ \Pi,\ \nu_t$	Timing association, shock loading, control coefficients, and residual
$h_u,\ h_d,\ W_t$	Upstream and downstream horizons, and the timing-regression controls
$\text{Spread}_t^j,\ r_t^j$	Lending spread and lending rate for segment j
$\pi_{t,t-12}^{DE}$	German annual HICP inflation

1. Introduction

1.1 Motivation

Between September 2019 and July 2022 the ECB held its deposit facility rate at minus 0.5 percent. Fourteen months later the rate stood at 4 percent, and by June 2025 it had been cut back to 2 percent. A path like that raises the question this thesis asks: through which channels does ECB policy reach the economy, and how long does its trace last? The question is old, but the post-COVID period sharpened it. Monetary transmission is not only about the average response of inflation or output. It is also about the channels through which financial conditions, bank intermediation, housing credit, and compensation-linked variables adjust over time. The Euro Area entered the pandemic after a long period of low rates, unconventional policy, and balance-sheet expansion. It then experienced extraordinary accommodation, followed by a rapid tightening cycle as inflation accelerated. In that environment, the same ECB announcement can move market prices inside the high-frequency window, shift lending margins over subsequent months, affect house-purchase credit growth, and leave a weaker or less consistent trace in compensation-linked proxies.

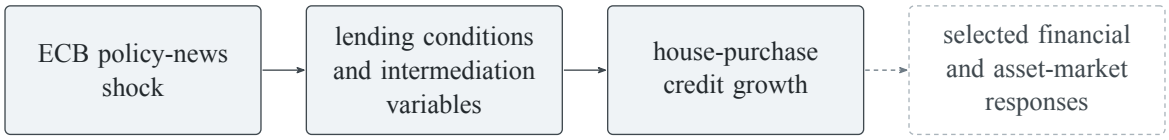
The stakes of that question are visible in the data this thesis uses. Between the first quarter of 2015 and the end of 2021, German house prices deflated by consumer prices rose by about 49 percent, while real compensation per employee in the euro area rose by about 4 percent over the whole decade to 2025. House-purchase lending in the euro area grew by about 3.3 percent a year through the low-rate years, slowed to about 2.2 percent a year once policy tightened, and dipped just below zero in April 2024. The gap between house prices and pay is the affordability backdrop against which the question is asked. This thesis does not measure who gained or lost access to housing. It measures the step before that: whether, and for how long, ECB policy news reaches housing credit and lending conditions at all, and whether it reaches them more clearly than it reaches pay.

This thesis studies that heterogeneous transmission. Its central argument is not that monetary policy mainly inflated assets or that broad real-economy transmission was absent. Those claims exceed what aggregate local projections can identify. The thesis instead asks whether ECB easing shocks, read against the post-COVID policy cycle, transmitted most clearly and

persistently through housing-credit and selected financial-intermediation variables, while evidence for compensation-linked real-economy transmission was weaker and less consistent.

The empirical contribution is therefore mechanism-focused. The required interpretation chain is:

Figure 1.1: Reduced-form transmission chain guiding the empirical interpretation. The diagram is a mechanism guide, not a structurally identified mediation model. Equity and asset-market responses are read alongside the chain under information-effect caution, not as a strict downstream consequence of housing-credit expansion.



Source: own illustration.

The framework is a high-frequency identified, dynamic impulse-response design. Its focus is interpretive. Housing-credit transmission and intermediation variables sit at the center of the thesis, while equity prices and broad financial indices are supporting evidence that must be read with information-effect caution.

1.2 Research Question

The thesis is organized around the following research question:

> *Did post-COVID ECB monetary easing transmit more strongly through housing-credit and financial-intermediation channels than through compensation-linked real-economy variables?*

The question is comparative, dynamic, and reduced-form. It concerns relative response strength and persistence across observable channels. Stronger transmission here means greater persistence, an accumulated response that keeps its sign and excludes zero through the longer horizons, together with sign consistency across horizons and samples. It does not mean significance at a single medium horizon alone. The question does not ask whether monetary policy caused household-level welfare changes, generated inequality, or shifted resources across population groups. Those questions require household, regional, or bank-level microdata that are outside the scope of the empirical design.

1.3 Hypotheses

Three hypotheses discipline the empirical interpretation.

Hypothesis 1 (Housing-credit persistence)**.**
Expansionary ECB policy-news shocks generated clear and persistent responses in housing-credit variables, especially house-purchase lending growth.

Hypothesis 2 (Compensation-linked attenuation)**.**
Compensation-linked real-economy variables responded less consistently and less persistently than the housing-credit block.

Hypothesis 3 (Differential financial-intermediation transmission)**.**
Expansionary monetary-policy shocks generated stronger and more persistent responses in housing-credit and financial-intermediation variables than in productive-credit and compensation-linked variables, indicating heterogeneous monetary transmission across sectors.

These hypotheses keep the thesis close to the evidence. Hypothesis 1 gives house-purchase lending growth its central role. Hypothesis 2 frames wages and labor-income proxies as a comparative block rather than as a welfare test. Hypothesis 3 replaces overbroad distributional language with a testable macro-financial transmission claim. Section 3.4 derives all three from the credit channel of monetary transmission and states, for each, the pattern in the response paths that would reject it.

1.4 Empirical Object

The thesis estimates monthly local-projection responses to high-frequency ECB target-factor surprises, aggregated to the month and signed so that positive values denote easing. Responses are estimated at horizons of 0, 1, 3, 6, 12, and 24 months and normalized to a one-standard-deviation surprise. Each response is estimated on all months available for it between 2005 and 2025, for statistical power. The label post-COVID denotes the motivating policy context that renews the salience of these channels, not a restriction of the estimation window.

The empirical object is a horizon-specific response profile to measured ECB policy news. The comparison spans house-purchase lending growth, pure-new mortgage flows, mortgage and NFC lending spreads, broad credit stocks, ECB liquidity-stock context, equity-market responses, labor expectations, and negotiated wage-pressure proxies. Quarterly house prices, compensation per employee, and Bank Lending Survey indicators are not interpolated into monthly responses. They remain context or robustness material when their observed frequency allows.

This object supports claims about differential transmission strength, response persistence, timing, and robustness. It does not identify household-level incidence, welfare effects, inequality, redistribution, exact structural mediation, or bank-specific treatment effects. That boundary is central to the thesis because the final inference is strongest when the estimand remains tied to observable repository outputs.

1.5 Contribution

The thesis makes five contributions.

1. It brings high-frequency ECB policy-news surprises into a monthly local-projection framework and traces their effects across housing credit, lending conditions, credit aggregates, asset prices, and compensation-linked variables over two years.

2. It makes house-purchase lending growth the central empirical variable and shows that it carries the clearest and most persistent response to policy news, a housing-credit result rather than a broad asset-market claim.

3. It measures financial intermediation directly, through lending margins over the policy rate, lending rates, credit aggregates, balance-sheet context, and timing evidence, and shows that the margin response is the most stable in the whole set.

4. It screens every announcement for information effects and reads the negative real DAX response through that screen, as mixed financial evidence rather than as a contradiction.

5. It states the claim boundary directly and reports its own uncertainty in full, including where the accumulated intervals depend on how dependence across horizons is handled, so that every claim can be traced to a reproducible output.

2. Literature Review

Chapter 1 set out the research question and the three hypotheses. This chapter reviews the literature they draw on, in the order the argument needs it. It starts with the theory of monetary transmission, moves to high-frequency identification and the information effects that complicate it, then to the evidence on housing credit and on macro-financial channels, and finally to the local-projection method. It closes with the gap the thesis fills.

2.1 Monetary Transmission Theory

The starting point is the credit-channel view of monetary policy. In the conventional interest-rate channel, a policy innovation affects short rates, the term structure, borrowing costs, asset prices, and expenditure. Credit-channel models add that policy also changes the external finance premium and borrower balance sheets, so that the effect of a policy shock depends on intermediation frictions and collateral conditions (Bernanke & Gertler, 1995). The financial accelerator extends this logic by showing how changes in borrower net worth amplify shocks through agency costs and credit access (Bernanke et al., 1999).

This framework is central for the thesis because it makes housing credit and bank intermediation first-order transmission objects rather than secondary controls. A policy-news shock can affect the price of credit through lending spreads, the quantity of credit through loan stocks and mortgage flows, and collateral-sensitive sectors through house-purchase lending. The mechanism need not imply that all real-economy variables respond with equal strength or persistence. Heterogeneous responses across lending margins, mortgage credit, broad credit, and compensation-linked variables are a natural feature of the theory.

The post-GFC and post-COVID Euro Area environment made these mechanisms more visible. Policy rates reached effective lower bounds, balance sheets expanded, forward guidance became central, and asset-purchase programs affected term premia, liquidity, duration risk, and portfolio rebalancing. In such an environment, monetary transmission is unlikely to be summarized by a single short-rate movement. It is better understood as a set of channel-specific

responses across market prices, bank balance sheets, lending conditions, mortgage credit, and income-linked variables.

Financial intermediaries provide the bridge between policy news and the housing-credit result. Adrian and Shin (2010) place intermediary leverage and balance-sheet capacity at the core of monetary economics, while Borio and Zhu (2012) emphasize the risk-taking channel. These accounts imply that monetary policy can affect the willingness and capacity of intermediaries to bear risk, price loans, and allocate credit. The thesis therefore treats lending spreads, rates, credit aggregates, and house-purchase lending growth as mechanism variables, not merely as auxiliary outcomes.

2.2 High-Frequency Identification and Information Effects

High-frequency identification addresses the timing problem in monetary economics. At monthly or quarterly frequency, observed policy changes are endogenous to inflation, slack, financial stress, and central-bank forecasts. Measuring market movements inside narrow policy-announcement windows helps isolate unexpected policy news (Gertler & Karadi, 2015; Kuttner, 2001). For the Euro Area, Altavilla et al. (2019) show that ECB announcements contain several dimensions of news, including target-rate, timing, forward-guidance, and asset-purchase components.

The same literature also explains why interpretation must be cautious. Announcement-window surprises are not automatically pure policy shocks. Nakamura and Steinsson (2018) emphasize that monetary announcements can reveal central-bank information about future fundamentals. Jarociński and Karadi (2020) develop this insight by distinguishing monetary policy shocks from central-bank information shocks. This matters directly for the thesis because an easing surprise can coincide with adverse macroeconomic information if the central bank signals weaker growth or financial stress.

This distinction matters most for equity evidence. An easing surprise can raise valuations through lower discount rates and at the same time lower them if it reveals weaker expected growth. The sign of an equity response to policy news is therefore ambiguous in theory. That is why this thesis reads equity evidence with information-effect caution rather than as direct evidence of asset-price appreciation, and why the housing-credit and lending-condition variables carry the stronger mechanism claim.

External-instrument work reinforces the same discipline. Designs that use policy surprises as instruments require a relevant bridge between the high-frequency surprise and the low-frequency treatment or response object (Mertens & Ravn, 2013; Stock & Watson, 2018). Instrument strength diagnostics such as F-statistics, partial R^2, event concentration, and jackknife sensitivity determine how far the treatment interpretation can be pushed (Stock & Yogo, 2005). The thesis therefore speaks of reduced-form responses to ECB policy-news shocks rather than fully purified structural treatment effects.

2.3 Housing-Credit Transmission

Housing is highly sensitive to monetary policy because it is credit-financed, collateral-intensive, and balance-sheet dependent. Mortgage rates affect user costs and borrowing constraints. Collateral values affect credit access. House-purchase lending connects financial conditions directly to housing-market participation. For these reasons, housing-credit variables deserve separate attention from broad household-credit aggregates.

The housing-credit-cycle literature supports this distinction. Jordà et al. (2015) link housing booms, leverage, and financial instability, showing that real estate credit is central to modern financial cycles. Jordà et al. (2016) further show that the growth of mortgage finance has transformed the relationship between banking systems, housing, and business cycles. These findings motivate the thesis' focus on house-purchase lending growth as the centerpiece variable.

A key empirical distinction is between stock-based credit growth and flow-based new lending. Stock-based house-purchase credit growth captures cumulative financing conditions, sustained pass-through, and medium-run credit accommodation. Flow-based pure-new mortgage lending captures immediate origination behavior and is more exposed to short-run volatility, sample length, and refinancing or reporting composition. A stronger response in lending growth than in pure new loans is therefore not a contradiction. It is a stock-flow distinction that helps interpret the mechanism.

The thesis' housing-credit claim is narrower than a house-price claim. Quarterly house prices are not interpolated into monthly responses, and the monthly model does not estimate household access outcomes, borrower incidence, or wealth distribution. The identifiable contribution is that house-purchase lending growth responds more clearly and persistently than compensation-linked proxies and broad productive-credit measures.

2.4 Macro-Financial and Asset-Price Transmission

Macro-financial transmission research emphasizes that policy news can affect market prices, intermediary balance sheets, liquidity conditions, and global financial conditions quickly (Adrian & Shin, 2010; Borio & Zhu, 2012; Rey, 2015). This literature is useful for the thesis because it motivates why financial and credit variables may react differently from compensation-linked variables. It does not by itself establish household incidence or distributional effects.

In a market-based financial system, policy news can move through discount rates, risk premia, collateral values, portfolio rebalancing, and bank risk-taking. These mechanisms may produce stronger responses in financial and credit variables than in compensation-linked variables, especially when wage adjustment is shaped by bargaining institutions, staggered contracts, expectations, and labor-market frictions (Taylor, 1980). The thesis uses this logic to motivate sectorally differentiated transmission rather than a broad asset-market narrative.

Equity-market responses require special caution. Asset prices are forward-looking and can react immediately, but their signs reflect multiple channels. An easing surprise can raise valuations through discount-rate effects while lowering them through adverse central-bank information about expected growth. The negative DAX response is therefore not hidden or forced into the housing-credit result. It is treated as evidence that financial-market responses are mixed and information-sensitive.

2.5 Local Projections and Dynamic Persistence

Local projections are well suited to the research question because they estimate impulse responses horizon by horizon (Jordà, 2005). A response that appears on impact and disappears by h3 has a different economic meaning from one that remains visible at h6, h12, or h24. Since the thesis is about relative persistence across channels, the path is more informative than a single impact coefficient.

The local-projection framework also clarifies the difference between peak and accumulated evidence. House-purchase lending growth may peak at a medium horizon and remain positive in accumulated terms, while compensation-linked proxies may move at shorter horizons and lose long-horizon precision. This comparison is exactly the reduced-form object the thesis can identify.

LP flexibility must be paired with careful inference. Overlapping horizons create dependence across residuals, and persistent macro-financial variables can make long-horizon inference fragile. HAC covariance estimates and bootstrap checks are therefore part of the evidence rather than an afterthought (Newey & West, 1987).

2.6 Research Gap

The literature provides strong tools for studying monetary transmission, but it often emphasizes immediate asset-price movements, broad macro aggregates, or average effects. Less attention is given to whether high-frequency ECB easing shocks transmit more clearly through housing-credit and financial-intermediation variables than through compensation-linked variables.

This thesis fills that gap by combining four elements. First, it uses high-frequency ECB policy-news shocks and monthly local projections. Second, it makes house-purchase lending growth the core empirical contribution. Third, it places lending spreads, credit aggregates, and timing evidence between the policy shock and the housing-credit response. Fourth, it treats market-price results, especially the DAX, as information-sensitive rather than as straightforward evidence of broad asset-price appreciation.

The result is a disciplined macro-financial transmission thesis. It asks where ECB easing shocks, read against the post-COVID policy cycle, remain most visible, shows that the housing-credit channel is the clearest and most persistent response surface, and keeps the interpretation within reduced-form identification boundaries. The next chapter turns this framing into a theory, a mechanism chain, three derived hypotheses, and an estimator.

3. Theoretical Framework and Methodology

Chapter 2 supplied the theory and the tools. This chapter puts them to work. It states the empirical objective, names the theory, derives the hypotheses from it, and then builds the estimator step by step: the shock, the screen for information effects, the relevance diagnostics, the local projections, the uncertainty layer, and the timing checks. It ends with what the design can and cannot say and with how every result is reproduced.

3.1 Empirical Objective

This thesis estimates how a set of macro-financial variables respond to unexpected ECB policy news. The shock is the monthly high-frequency target surprise. It is signed so that a positive value is an easing surprise. The variables cover housing credit, lending conditions, financial-market prices, the ECB balance sheet, broad credit stocks, labor expectations, and compensation-linked wage pressure.

The question is comparative. It asks which of these channels shows the clearest and most persistent response, not whether a single channel moves in isolation. The object of interest is therefore a response path, measured at several horizons, rather than one impact number. Signs, accumulated responses, confidence bands, and stability checks are read together.

The estimates are reduced-form. They describe dynamic responses to policy news. They are not a structural welfare model, and they do not measure household-level outcomes. That boundary is set here, and it holds through the results and the conclusion.

3.2 Theoretical Foundation: The Credit Channel

The theory behind the thesis is the credit channel of monetary policy transmission, set out by Bernanke and Gertler (1995) and given a formal quantitative model in the financial accelerator of Bernanke et al. (1999). The credit channel starts from one friction. Lenders know less about borrowers than borrowers know about themselves, so external funds cost more than internal funds. The gap is the external finance premium. Monetary policy moves this premium as well

as the risk-free rate, and the premium is what makes the response of spending to a policy change larger and more uneven across sectors than the interest-rate channel alone would produce.

The theory has two strands. The balance-sheet channel says that policy changes borrower net worth and collateral values. Easier policy raises asset prices and cash flows, strengthens balance sheets, and lowers the premium a borrower has to pay, so credit-financed spending expands. The bank-lending channel says that policy changes the supply of loans that banks are willing to make. Easier policy improves bank funding conditions and balance-sheet capacity, so banks price loans more generously and extend more of them. Adrian and Shin (2010) and Borio and Zhu (2012) extend the second strand to the intermediary itself. Policy changes the leverage, funding costs, and risk appetite of banks, and that shows up first in lending margins and lending standards, before it shows up in credit volumes or in the real economy.

Two implications of the theory matter for this thesis. First, the effect of a policy shock is uneven across sectors. It is largest where spending is credit-financed and secured on collateral, because that is where the external finance premium and the collateral constraint bind. Housing is the leading case. Bernanke and Gertler (1995) point to it as the purchase where the conventional interest-rate channel struggles most, since a temporary move in short rates has a large effect on demand for a long-lived asset, and the credit channel is their explanation. Second, the effect on income runs through activity, not directly. Policy changes financial conditions, financial conditions change credit-financed spending, and only then do output, employment, and pay adjust. Pay adjusts slowly because nominal wages are set in multi-period contracts that overlap across firms and sectors, so at any point in time most wages cannot respond to news (Taylor, 1980). Negotiated wages, which the compensation-linked block in this thesis measures, are the clearest case of such contracts.

3.3 Mechanism Framework

The credit channel is read through a housing-credit transmission chain.

$$\begin{aligned}\text{ECB policy news} &\rightarrow \text{intermediation variables} \\ &\rightarrow \text{housing-credit expansion} \rightarrow \text{selected financial responses.}\end{aligned} \tag{3.1}$$

The first link is the ECB policy surprise. The second link is the set of intermediation variables, mainly lending spreads, lending rates, broad credit, the balance sheet, and bank timing evidence.

The third link is housing credit, measured first by house-purchase lending growth and then by pure-new mortgage flows. Equity and other financial prices sit alongside this chain and help interpret it. They do not carry the thesis on their own.

The chain is not a structural mediation model. All of the variables respond to the same shock, and monthly aggregate data cannot separate the decision of one bank from the response of one household. What the chain does provide is a disciplined way to read timing. If policy news reaches housing credit through intermediation, the lending variables should move in a way that makes the housing-credit response believable. The compensation-linked block is the real-economy comparison.

3.4 Derivation of the Hypotheses

The research question stated in Chapter 1 asks whether post-COVID ECB monetary easing transmitted more strongly through housing-credit and financial-intermediation channels than through compensation-linked real-economy variables. The credit channel gives a direct answer, and the three hypotheses follow from it. They are restated here in the order of the chain in Equation (3.1), so that each can be read against the part of the theory that produces it.

The first hypothesis follows from the balance-sheet channel. Housing is the largest credit-financed and collateral-secured purchase most households make. An easing surprise lowers the external finance premium on mortgage borrowing and raises the collateral value that secures it, and both effects work in the same direction. Because the stock of housing credit adjusts through new lending that is repaid over years, the effect on the growth of that stock should build over several months and last, not appear and vanish at impact. This is Hypothesis 1: expansionary ECB policy-news shocks generated clear and persistent responses in housing-credit variables, especially house-purchase lending growth. The hypothesis is rejected if the accumulated response of house-purchase lending growth includes zero, or changes sign, at the twelve- and twenty-four-month horizons.

The second hypothesis follows from the sequencing of the channel together with staggered wage setting. Compensation sits at the end of the chain. It moves only after credit-financed spending has changed activity, and negotiated wages respond to that change only as agreements come up for renewal. Within a two-year window the monthly response of compensation-linked variables should therefore be later, smaller, and less stable than the response of housing credit.

That is Hypothesis 2: compensation-linked real-economy variables responded less consistently and less persistently than the housing-credit block. The hypothesis is rejected if the accumulated compensation-linked responses keep their sign and exclude zero at the long horizons as reliably as housing credit does. It is a comparative statement, not a claim that income effects are absent.

The third hypothesis follows from the bank-lending channel and the intermediary view. Banks respond to policy first through the price of credit, that is the margin between their lending rates and their funding rate, and they expand lending first where collateral makes it cheapest to extend. Household mortgage lending is secured on real estate. Lending to firms depends on investment plans and on firm balance sheets that adjust more slowly. The theory therefore predicts that intermediation variables and housing credit respond earlier and more persistently than productive-credit quantities and compensation-linked variables. Hypothesis 3 puts this as follows: expansionary monetary-policy shocks generated stronger and more persistent responses in housing-credit and financial-intermediation variables than in productive-credit and compensation-linked variables, indicating heterogeneous monetary transmission across sectors. The hypothesis is rejected if broad NFC credit responds as strongly and as durably as housing credit and the lending spreads. The timing evidence in Section 3.11 is descriptive material on that ordering, not a test of it.

All three hypotheses are testable with the response paths defined in the rest of this chapter. Stronger transmission is measured as defined in Chapter 1: an accumulated response that keeps its sign and excludes zero through the longer horizons, together with sign consistency across horizons and samples. Chapter 5 reports the tests and Table 5.5 collects the verdicts.

3.5 The Identification Problem

Monetary policy is not random. The ECB reacts to inflation, to slack, to financial stress, and to its own forecasts. A simple regression of any outcome on a policy-rate change therefore mixes three things. It mixes the policy action, the private sector's anticipation of that action, and the information the central bank reveals when it acts.

High-frequency identification narrows the timing. It measures the change in yields or asset prices inside a short window around each ECB announcement (Altavilla et al., 2019; Gertler & Karadi, 2015; Kuttner, 2001; Nakamura & Steinsson, 2018). A movement in that window is hard to explain with slow-moving macro data, so it is a cleaner read on what markets did not

expect. This does not solve everything. An announcement can also reveal the ECB's own view of the economy, which is the information effect (Jarociński & Karadi, 2020). For that reason the surprise is used as measured policy news. It is not treated as a fully clean structural shock.

3.6 Constructing the Shock Series

Three frequencies meet in this thesis, and the chain between them is stated here once. The surprises are measured intraday. The ECB's Euro Area Monetary Policy Event-Study Database records how OIS rates, sovereign yields, and equity prices change inside a short window around each Governing Council communication, from shortly before the press release or the press conference begins to shortly after it ends (Altavilla et al., 2019; European Central Bank, n.d.-d). Altavilla et al. (2019) compress those window changes into a small number of factors. This thesis uses their target factor: the component measured in the press-release window and normalised to move the one-month OIS rate one for one, which makes it the surprise in the rate decision itself. The factor values come from the updated vintage of 30 October 2025 (Ragusa, n.d.), so one target-factor value exists per announcement. Estimation runs at the monthly frequency. The announcement-level values are summed to months as set out below, and every response variable, every control, and every regression in Chapters 5 and 6 is monthly. Series observed only at quarter ends are never disaggregated to reach that frequency, and Chapter 4 explains how they are handled instead.

Let z_e^f be the high-frequency surprise for factor family f at event e. The monthly value sums the events in the month.

$$Z_m^f = \sum_{e \in \mathcal{E}(m)} z_e^f, \tag{3.2}$$

Here $\mathcal{E}(m)$ is the set of ECB events in month m. Summing keeps the total news that arrived in the month, which matters when a month contains more than one relevant communication. Months with no event take a value of zero inside the sample coverage. Two events have no target-factor value in the surprise data, the inter-meeting cut of 8 October 2008 and the meeting of 6 November 2008. October 2008 therefore carries only the 2 October surprise, and November 2008 has no shock value and drops out of the regressions. Months after the source series ends are dropped.

The shock used in the thesis flips the sign of the target factor.

$$\text{Shock}_m = -Z_m^{target}, \tag{3.3}$$

After the flip a positive value is an easing surprise and a negative value a tightening surprise. The reference point is what markets expected just before the communication, not the previous level of the rate. A cut that markets had fully priced is close to zero, and a hike smaller than the one they had priced is an easing surprise. The size of the value is the size of the revision in short-term rate expectations inside the window. In the estimation sample the monthly series has a standard deviation of 2.41 and ranges from −21.1 to 13.7, so a few announcements carry much of the variation, a feature Chapter 6 returns to. The timing, forward-guidance, QE, and weighted-composite factors are kept for the robustness chapter.

3.7 Screening for Information Effects

Not every announcement is a clean monetary signal. The screen compares three signs at each event: the rate factor, the six-month OIS change inside the announcement window, and the STOXX50 change in the same window. The rate factor is the target factor, or the timing factor when the event has no target surprise, and the timing factor also stands in for the OIS change when that window is missing. When the yield move and the equity move have opposite signs, the event looks like a cleaner monetary surprise. When they share a sign the event is marked as a possible information shock, when the rate factor and the yield move disagree it is marked as a sign conflict, and zero or missing market responses are marked as ambiguous. Everything except the clean class counts as contaminated. In the monthly identification window, January 2005 to October 2025, 76 events are clean and 132 are contaminated or ambiguous.

The baseline keeps the full target shock. The clean-event and contaminated-event series are then used to test how far the results depend on the noisy events. The negative DAX response reported later is read under this same caution. An easing surprise can carry bad news about the economy, and equity markets react to both parts.

The screen has a limit that should be stated. Its sign rule is a necessary condition for a pure monetary surprise, not a sufficient one. Yields and equities moving in the same direction is inconsistent with a policy shock alone, so the rule rejects such events with some confidence. The

converse does not hold. Yields falling while equities rise is what an easing surprise produces, but it is also what any unrelated piece of good news arriving inside the window would produce, so an event that passes the screen may still carry information that has nothing to do with the policy decision. The 76 clean events are therefore events the rule could not reject, not events shown to be pure. Establishing purity would require checking each window against the news flow of the day, which lies outside this thesis. Two features of the design limit the damage. The baseline keeps every event, so no headline result rests on the classification, and the clean and contaminated series enter only as a robustness comparison in Chapter 6. If the clean sample is itself partly contaminated, that comparison understates rather than overstates how far the two groups differ.

3.8 Relevance and the Interpretation Boundary

A natural question is whether the surprise can instrument a specific low-frequency policy variable, such as the change in the deposit facility rate, the shadow rate, or ECB assets (Mertens & Ravn, 2013; Stock & Watson, 2018). A first-stage tournament tests this across every candidate pairing, 45 in all. Each first stage regresses the candidate policy variable on the surprise, two lags of the policy variable, and the current and lagged monthly euro-area inflation rate. It records the conventional first-stage F-statistic from the restricted and unrestricted residual sums of squares, a HAC t-statistic with three lags, and the partial R^2, the share of the residual variance of the policy variable that the surprise explains (Stock & Yogo, 2005). Two sensitivity checks follow. A jackknife drops the five months with the largest absolute surprises and re-estimates, and rolling 60-month windows, moved one month at a time, track whether relevance is steady across the sample and across the four policy eras defined in Section 6.3.

The best bridge is the end-of-month change in the deposit facility rate on the easing-signed target surprise, with $F = 17.07$ and partial $R^2 = 0.066$. That is real full-sample relevance, not a weak-instrument collapse. The problem is stability. The relevance is carried by a small number of large events, and Chapter 6 shows how quickly it weakens once those events are removed. It is not steady enough to support a strong claim about one treatment variable. The thesis takes the honest route. It estimates responses to the policy-news shock itself, and it does not upgrade the surprise into an instrument for the DFR or the balance sheet.

3.9 Local Projections

The responses are estimated with local projections (Jordà, 2005). One regression is run for each response variable, each horizon, and each shock.

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \text{Shock}_t + \Gamma_h X_t + \varepsilon_{t+h}, \qquad h \in \{0, 1, 3, 6, 12, 24\}. \tag{3.4}$$

Here y is the response variable, t is the month of the ECB communication, h counts months after it, and ε_{t+h} is the regression error. The control vector holds six lagged values,

$$X_t = (y_{t-1},\ y_{t-2},\ \pi_{t-1},\ \pi_{t-2},\ r_{t-1},\ r_{t-2}), \tag{3.5}$$

where π is month-on-month euro-area inflation and r is the monthly-average deposit facility rate. The subscript on X_t marks the information set at the time of the shock. Every element of it is dated before month t. The coefficient β_h is the response of y at horizon h to a unit surprise, and it is the only coefficient that is interpreted. The next five subsections justify each part of the equation: the dependent variable and its base month, the dating of the shock, the dating of the controls, the treatment of levels and rates, and the choice of local projections over a VAR.

3.9.1 The Dependent Variable and Its Base Month

The left side is a long difference. It measures how far y has moved h months after the communication, relative to its value in the month before it. The base is y_{t-1} rather than y_t for a timing reason. The communication falls inside month t. The monthly observations of the response variables are month-end values or averages over the month, so y_t is recorded after the news has arrived and may already contain part of the response to it. The last observation that is certain to predate the shock is y_{t-1}. Measuring from there keeps the whole response inside the dependent variable, including the part that occurs within the month of the announcement, which is what the regression at $h = 0$ estimates.

Because y_{t-1} is itself a control, the choice of base among the pre-shock values does not affect $\widehat{\beta}_h$. Subtracting y_{t-1} from the left side only shifts the coefficient on that control by one, and a regression of the level y_{t+h} on the same right-hand side returns the same shock coefficient to machine precision in every one of the 60 baseline cells. The base must predate the shock, and

beyond that it is a choice of units. The long difference makes the left side a change, which is the unit in which a response path should be read. Using y_t as the base would be different in kind. It would hold a post-shock value fixed on the right-hand side, which is the problem Section 3.9.3 describes, and it would force the impact response to zero by construction.

3.9.2 The Shock and Its Dating

The shock is dated t, the month in which the communication took place, and it enters once. It is not lagged. The surprise is realised at a known moment inside month t, nothing about it is observed before that moment, and the response variables from month t onward are the first that can react to it. Pairing $y_{t+h} - y_{t-1}$ with Shock_{t-1} would date the news one month before it existed. No lags of the shock are added either. A local projection at horizon h already measures the effect of the month-t surprise over the following h months, so the dynamics come from varying h across the six regressions, not from a distributed lag inside any one of them. This is the design of Jordà (2005), and it is what makes the coefficient at each horizon readable as a point on the response path. Earlier surprises also affect y_{t+h}, but they do not need to be controlled for, because they are nearly uncorrelated with the month-t surprise. Each surprise is measured against the expectations that prevailed just before its own announcement, and the first-order autocorrelation of the monthly shock in the estimation sample is -0.09. Their effect therefore sits in the error term without biasing β_h.

3.9.3 The Controls and Their Dating

The controls serve one purpose. They remove from $y_{t+h} - y_{t-1}$ the part that was predictable before the shock arrived, so that β_h is estimated from the variation that remains. Each element of X_t is dated $t - 1$ or $t - 2$, and that dating is deliberate. A control must be correlated with the outcome and must not itself be moved by the shock. Any variable dated t, $t + 1$, or later fails the second condition, because the surprise can move it. Conditioning on such a variable would hold fixed part of the channel through which the shock works, and β_h would then measure only the remainder. Angrist and Pischke (2009) call variables of this kind bad controls, and the rule is general: the controls in a response regression must be fixed before the treatment. The variation in y_{t+h} that is caused by events after month t is therefore left in the error term. That widens the standard errors, which is why the uncertainty layer in Section 3.10 is extensive, but

it does not bias β_h, because the surprise is measured against the expectations held just before the communication and is uncorrelated with news that had not yet arrived.

Each pair of lags does a specific job. The two lags of the response itself absorb its own momentum, so that a movement already under way before the announcement is not credited to it. They also make the inference robust to persistence. Montiel Olea and Plagborg-Møller (2021) show that a local projection augmented with lags of all variables in the system supports inference with standard critical values whether the series is stationary or close to a unit root, provided the horizon is short relative to the sample, which matters here because the log levels in the response set are highly persistent. The two lags used here follow that logic without meeting every condition of their result. The two lags of inflation hold fixed the state of the economy to which the ECB was responding, and the two lags of the deposit facility rate hold fixed the policy stance, so that the same surprise is compared across months in which rates stood at minus one half of a percent and at four percent. Together the inflation and policy-rate lags absorb the predictable component of the policy decision, the part a reaction function would produce. Two lags are used throughout. Chapter 6 explains why a longer lag grid is not run and which checks take its place.

3.9.4 Levels, Rates, and Growth

The response variables come in two forms. Spreads, growth rates, and survey balances are rates, measured in percentage points or index points. The DAX, the credit stocks, the ECB balance sheet, and new mortgage loans are levels and enter in logs. The long difference treats both consistently. For a rate it is the change in the rate over h months, in percentage points. For a log level it is the log change, which is close to the percentage change in the level. The left side is therefore always a change, never a level, and each response is read in its own units. The comparison across responses in Chapter 5 is of sign, persistence, and stability, not of size.

The two types also differ in how they move over time, and the right side has to allow for that. Log levels are highly persistent and trend upward. Spreads and growth rates revert. The lags of the response carry this difference into the regression. Because y_{t-1} and y_{t-2} together determine the last pre-shock change $y_{t-1} - y_{t-2}$, conditioning on the two lagged values is the same as conditioning on the last pre-shock level and the last pre-shock change. For a log index the regression holds fixed where the index stood and how fast it was growing when the news arrived. For a rate it holds fixed the level of the rate and its last move. The persistence of the error

term also differs across the two types, and nothing in the specification forces it to be the same. The HAC bandwidth and the bootstraps of Section 3.10 are computed response by response from each regression's own residuals, and Table 6.3 reports what those residuals look like.

3.9.5 Why Local Projections

Local projections are used instead of a VAR for three reasons. The interest is in the path of each response rather than in the full system of dynamic interactions. The horizons are irregular, and a VAR would have to be iterated out to each of them from a single lag structure. And a local projection stays robust when the longer dynamics are hard to specify, because each horizon is estimated directly rather than by iterating a one-step model forward. The cost is that each regression spans h months, so neighbouring observations of the dependent variable overlap and the errors are serially correlated by construction. Section 3.10 is built around that cost.

3.9.6 Scaling and Accumulation

Each coefficient is scaled to a one-standard-deviation shock, with the standard deviation taken over the estimation sample of that regression. A cell is estimated only when at least 48 months are available.

$$\widehat{\mathrm{IRF}}_h = \widehat{\beta}_h \widehat{\sigma}_{\mathrm{Shock}}. \tag{3.6}$$

The scaled coefficients give the impulse response at each horizon in the response's own units. Their running sum over the reported horizons gives the accumulated response,

$$\widehat{\mathrm{CIRF}}(H) = \sum_{h \in \mathscr{H}:\, h \le H} \widehat{\mathrm{IRF}}_h, \qquad \mathscr{H} = \{0, 1, 3, 6, 12, 24\}, \tag{3.7}$$

which is the main measure of how long an effect lasts. It is a persistence index rather than a cumulative multiplier. It sums the six reported horizons, not every month between them, so its level has no reading as a total effect. What it does is reward a response that keeps its sign across horizons and penalise one that changes sign. It rises as long as the single-horizon responses stay positive and falls when they turn negative. A fall in the accumulated response between two horizons is therefore read in Chapter 5 as a partial reversal of the earlier response, and a response is called persistent only if its accumulated value keeps its sign and excludes zero at the longer horizons, as Chapter 1 defined the term.

3.10 Uncertainty

The baseline standard errors are HAC Newey–West with bandwidth $h + 1$ (Newey & West, 1987). The bandwidth grows with the horizon because a local projection at horizon h overlaps its neighbors, so the residuals are serially correlated by construction.

The accumulated responses need a second step. Their HAC intervals add the horizon-level variances as if the horizon estimates were independent,

$$\widehat{\sigma}_{\mathrm{CIRF}}(H) = \Big(\sum_{h \in \mathscr{H}:\, h \le H} \widehat{\sigma}_h^2 \Big)^{1/2}, \tag{3.8}$$

where $\widehat{\sigma}_h$ is the HAC standard error of the scaled coefficient at horizon h. That is an approximation. Adjacent projections share residuals, and Chapter 6 measures how strongly. Three bootstraps therefore sit next to the HAC intervals. A fixed-design residual bootstrap with circular blocks of length $\max(3, h + 1)$ gives a second interval for every horizon-level coefficient of the target shock. A joint moving-block bootstrap resamples the residuals of all six horizons together, in blocks of twelve months, and so keeps the dependence across horizons when it builds the accumulated path. A wild bootstrap does the same with random sign flips instead of blocks. The wild bootstrap keeps the dependence across horizons but not across months. Each uses 399 replications and percentile intervals. Where the HAC interval and the joint bootstraps disagree on an accumulated response, Chapter 5 says so, and Table 6.4 reports both.

The bootstraps rest on properties of the residuals that can be checked, and Chapter 6 reports those checks in Table 6.3 for every canonical response at every horizon. Serial correlation up to lag h is expected by construction and is what the HAC bandwidth covers, so the relevant test is for correlation beyond the bandwidth (Ljung & Box, 1978). Heteroskedasticity, unconditional or conditional, is what the wild bootstrap is designed to tolerate, so its presence is the reason the wild interval is reported alongside the moving-block interval (Breusch & Pagan, 1979; Engle, 1982; Koenker, 1981). Non-normality is the reason every bootstrap interval is a percentile interval and none relies on a normal approximation to the bootstrap distribution (Jarque & Bera, 1980).

3.11 Sequential Timing

The mechanism chain suggests an order, with lending conditions adjusting before housing credit. Monthly aggregates cannot test that order directly, because every variable responds to the same shock within the same month. The sequential layer therefore asks a narrower question. It fixes an upstream variable u at an early horizon h_u and a downstream variable d at a later horizon h_d, and asks whether the early move carries information about the later one once the shock itself is held fixed:

$$d_{t+h_d} - d_{t-1} = a + \lambda(u_{t+h_u} - u_{t-1}) + \theta \text{Shock}_t + \Pi W_t + \nu_t. \tag{3.9}$$

Here W_t holds the first and second lags of the upstream variable, the downstream variable, inflation, and the deposit facility rate, dated as in Section 3.9.3. Unlike those controls, the upstream change $u_{t+h_u} - u_{t-1}$ is dated after the shock. That is intended. It is the variable whose association with the later downstream move is being measured, not a control, which is why λ is not given a causal reading. The coefficient λ is a conditional association between an early upstream move and a later downstream move. The order of the horizons is imposed by the design, so a positive and precise λ says that the two moves line up in time, not that one causes the other. A negative λ between a price and a quantity is the ordinary sign of a demand relation and is read as such. The peak horizons of the individual responses, reported in the appendix, are the second piece of timing evidence.

3.12 What the Model Can and Cannot Say

The model gives reduced-form dynamic evidence under imperfect identification. It shows where ECB policy news leaves the clearest and most persistent mark, and it compares the financial channels with a real-economy block. It is a study of heterogeneous transmission and financial propagation. It is not a welfare analysis, an inequality decomposition, or a study of household redistribution. The results chapter and the conclusion stay inside this line.

3.13 Reproducibility

The entire empirical layer is reproducible from a single pipeline. One command reruns the data build, the shock construction, the local projections, the uncertainty checks, and a final audit, and every figure and every numerical table in this thesis is written by that pipeline, while the three qualitative tables are typeset from the same outputs. A second script reproduces every table of the thesis in thesis order and numbering from the same outputs, and Section A.5 lists the canonical output files behind the reported evidence. The next chapter sets out the data and the proxy choices. The robustness chapter later returns to event contamination, historical exclusions, rolling and recursive windows, alternative shock factors, a raw-policy-rate comparison, and descriptive regimes.

4. Data

Chapter 3 defined what is estimated. This chapter describes the data it is estimated on: where each series comes from, which proxies stand in for the concepts the hypotheses name and why, how frequency is handled, how the series are transformed, and what the final sample looks like.

4.1 Data Architecture

The dataset is a monthly euro-area and Germany panel. It is built from high-frequency ECB surprise files (European Central Bank, n.d.-d; Ragusa, n.d.), ECB monetary and banking series (European Central Bank, n.d.-b), Eurostat indicators (Eurostat, n.d.-a), market data, and a set of official monthly proxy candidates from the same two portals. Table A.2 in the appendix lists every series with its code and access date. The estimation panel holds 250 monthly observations from January 2005 through October 2025, dated at month ends, across 171 variables. Two build steps produce it: one constructs the shock series from the announcement-window surprises, and one assembles the panel from the source series.

The full rebuild, from the shocks through the final audit, runs from the single pipeline set out in Chapter 3. Source manifests, coverage tables, and diagnostics form the factual reference layer, and every number in this chapter traces back to them.

4.2 Variable Selection Logic

A variable enters the panel only if it clears four rules. It has to trace to an official source. It has to be reproducible from the build scripts. It must be estimable at monthly frequency, or observed at quarter ends. And it has to be narrower than the thesis claim. The last rule does the most work. A mortgage-credit proxy is not a house-price measure, and a wage tracker is not total compensation. The administered policy rate is a different object again from the high-frequency shock. If each proxy stays narrower than the claim, the measurement cannot quietly become the result the estimates are meant to produce.

4.2.1 Monetary Policy and Liquidity Variables

The policy shock is the easing-signed monthly target surprise, Shock_t. It is the monthly sum of ECB target-factor announcement-window surprises, signed so that a positive value is an easing surprise. Its construction and sign convention are set in Chapter 3. The timing, forward-guidance, QE, and weighted-composite factors enter as robustness shocks. The QE factor starts later, in 2014-01, because that is when the source series begins.

The administered policy rate enters separately from the shock. The ECB deposit facility rate (European Central Bank, n.d.-c) is kept both as a month-end series and as a monthly-average series. First differences of the rate support bridge diagnostics. The monthly average feeds the local projection controls and the lending spreads. The Wu-Xia euro-area shadow rate covers 2005-01 through 2022-08 (Wu & Xia, 2016, 2020, n.d.). It is not extended, because that is where the source series ends. ECB asset stocks (European Central Bank, n.d.-a) arrive weekly. They are converted to month-end and monthly-average series, then transformed into logs, differences, and scaled variants.

4.2.2 Lending Conditions and Credit Quantities

This is the intermediation block, the link between policy news and the financial responses. Mortgage and NFC lending rates come from the ECB's interest-rate statistics. The main lending-condition variables subtract the monthly-average deposit facility rate from each lending rate, which gives the mortgage and NFC lending spreads defined in Equation (4.2). The spreads measure the borrower-facing margin over the policy-rate floor. That margin is what a bank sets, not what the central bank administers.

Broad credit enters as log household and log NFC loan stocks. Household loans are read as background credit rather than a targeted response. Housing finance is measured two ways. The canonical monthly proxy is the annual growth rate of ECB lending for house purchase, available across the full sample. The second measure is the log of pure new house-purchase loans, a flow series that begins in August 2017 and enters as robustness. ECB Bank Lending Survey variables are observed at quarter ends only. They give institutional context on credit standards and loan demand, not a monthly response.

4.2.3 Market, Housing-Credit, and Compensation Variables

The market-price response is the real DAX, the DAX close (Yahoo Finance, n.d.) deflated with German HICP (Eurostat, n.d.-b) and taken in logs. Quarterly German house prices (Bank for International Settlements, n.d.) are kept in real log form at their observed quarter-end dates. They are not converted into a monthly response, so the house-price series stays a quarter-end reference rather than an estimated path.

The canonical compensation response is the real ECB Wage Tracker excluding one-off payments, the nominal tracker minus EA20 annual HICP inflation, and it begins in January 2013. Headline, unsmoothed, moving-average, and momentum variants are robustness measures. Quarterly compensation per employee is kept in real log form, again at quarter ends only. A monthly robustness partner comes from the Eurostat STS German industry wage bill. Four more series enter as indirect labor-pressure and real-activity diagnostics: employment expectations from ECB and Eurostat sources, inverse unemployment tightness, the ECB Consumer Expectations Survey, and German retail volume. They frame the compensation block. They do not substitute for it.

4.3 Frequency Governance

Frequency is treated as a hard constraint, not a nuisance. Monthly estimation needs monthly variation. Some of the most conceptually direct series are observed only at quarter ends. Each such level series carries 83 observations, from 2005-03-31 to 2025-09-30, on quarter-end months alone.

The shock sits at the other end of the frequency ladder. It is measured intraday, one value per announcement, and summed to months before it enters any regression, as Section 3.6 sets out. The monthly frequency is therefore the meeting point. The shock is aggregated up to it, and the quarterly series are never disaggregated down to it.

Quarterly interpolation is rejected. Filling the missing months would create artificial smoothness and would spread a single quarterly move mechanically across three months. That would manufacture the very timing the thesis sets out to measure. So the quarterly series are left as observed. The cost is direct. Taking a monthly log difference of a quarter-end series returns

nothing usable. The processed coverage confirms it. The monthly log differences of the quarterly house-price, compensation, and Bank Lending Survey series contain zero observations.

The answer to this constraint is to estimate on narrower series that are genuinely monthly. House-purchase lending growth replaces mortgage finance, and the wage-tracker variables replace negotiated wage pressure. Both are less direct than the quarterly measures, but they are observed every month, which is what the estimator needs. The quarterly series stay in the panel as context, not as responses.

4.4 Proxy Validation Framework

Proxy selection is not left to judgment alone. A scoring framework in the pipeline rates each candidate and sorts it into one class. The classes are canonical, robustness, indirect, diagnostic, rejected, and unavailable.

Every candidate, housing and compensation alike, first passes a common scorecard. It is a weighted sum of ten sub-scores, each on a one-to-five scale. Four describe the data: economic interpretability, monthly continuity, missingness, and the realism of the series' volatility. Six describe the candidate's own response to the target surprise: sign stability across horizons, persistence, correlation with the quarterly concept the proxy stands in for, sign stability across subperiods, sensitivity to the clean-event screen, and consistency across regimes. Interpretability and continuity carry the largest weights. On the housing side the outcome keeps house-purchase lending growth as canonical, pure-new house-purchase loans as robustness, and broad household loans as general credit.

Compensation proxies then face a second, longer scorecard of fourteen weighted sub-scores. On the data side it rates economic interpretability, frequency integrity, revision risk, stationarity, and the signal-to-noise ratio. On the response side it rates shock sensitivity at six months, sign consistency across horizons, cumulative persistence, the share of horizons with HAC significance, clean-event sensitivity, rolling- and recursive-window stability, regime consistency, and bootstrap sign stability. Here the mechanical scoring and the final choice part ways. The scorecard ranks the Eurostat German industry wage bill first, with a score of 157. The ECB Wage Tracker excluding one-offs ranks third, at 141. The selection still keeps the wage tracker as the primary canonical compensation proxy and moves the German wage bill to a secondary robustness proxy. The reason is conceptual, not statistical. The wage tracker is the most interpretable

monthly wage-pressure measure, and the euro-area negotiated-wage concept fits the thesis better than a narrower national industry wage bill. No composite index is selected. Employment expectations and inverse unemployment tightness stay as indirect diagnostics.

The appendix reports the compensation proxy ranking in Figure A.1. It supports the selection and is not used as a separate results claim.

4.5 Transformation Pipeline

The transformations are uniform across the panel. Positive-valued stocks and indexes are logged. First differences subtract the previous monthly observation. Year-on-year growth uses twelve-month log differences. Real variables deflate the nominal series with the matching HICP index. The ECB wage trackers arrive as nominal annual growth rates, so the real versions subtract EA20 annual HICP inflation. The German industry wage-bill proxy follows the same real-growth rule,

$$100\left[\log(W_t) - \log(W_{t-12})\right] - \pi^{DE}_{t,t-12}, \tag{4.1}$$

where W_t is the monthly wage-bill index and $\pi^{DE}_{t,t-12}$ is German annual inflation.

The lending spreads are the borrower rate minus the policy floor,

$$\text{Spread}^j_t = r^j_t - \text{DFR}^{mavg}_t, \tag{4.2}$$

where j denotes mortgage or NFC lending. Response variables are left in their natural units and are not standardized. Only the shock response is normalized in the local projection output, which keeps every response readable on its own scale.

4.6 Final Sample Construction

Coverage across the 250-row sample is not uniform, and the gaps are handled by response-specific estimation samples rather than imputation. Each local projection uses the months where its own variables are present. Four series set the binding edges. The target shock is nearly complete across the sample. The Wu-Xia shadow rate ends in 2022-08. The pure-new house-purchase loans begin in 2017-08. The wage tracker begins in 2013-01.

Event filtering sits on top of the same panel. The screen in Chapter 3 sorts the events into clean, contaminated, and ambiguous samples, and Chapter 6 reports the full event breakdown. COVID, crisis-window, QE-launch, extreme-shock, rolling, recursive, and regime variants are robustness devices, not the baseline.

4.7 Descriptive Statistics and Correlations

Table 4.1 reports the shock and the ten canonical response variables. Three features matter for the estimation that follows. The easing surprise has a mean close to zero, which is what a measure of unexpected news should look like, and a standard deviation of 2.41 with a range from −21.1 to 13.7, so a few announcements dominate the variation. That concentration is the same feature the relevance diagnostics pick up in Chapter 6. House-purchase lending growth averages 3.62 percent a year and moves over a wide band, from −0.56 to 12.17, so there is real variation to explain. The lending spreads average slightly above two percentage points over the deposit rate, and the mortgage spread turns marginally negative at its minimum, which occurs in May 2024, when the deposit facility rate stood at 4 percent.

The differing sample sizes are the visible cost of the frequency rule set out above. The shock, housing credit, the spreads, broad credit, the DAX, ECB assets, and employment expectations are observed in all or nearly all 250 months. The wage tracker has 154 observations and the pure-new-loan series only 99, because both start later than the panel. The wage-tracker gap explains why the compensation-linked estimates are less precise at long horizons, and the pure-new-loan gap explains why that series stays a robustness surface rather than the canonical housing proxy.

Table 4.2 reports the pairwise correlations. The first column is the informative one for identification. The easing surprise is essentially uncorrelated with the contemporaneous level of every response, with coefficients between −0.14 and 0.13, and only two of them reach the five percent level. That is the expected pattern for a high-frequency surprise. It carries news rather than tracking the slow-moving state of the economy, so the dynamic responses in Chapter 5 are not driven by a mechanical contemporaneous association.

The correlations among the responses are much stronger and confirm that the blocks behave as blocks. The two lending spreads move almost together at 0.91, house-purchase lending growth and pure-new loans correlate at 0.69, and the real DAX and ECB assets correlate at 0.82. Housing-credit growth is negatively correlated with broad NFC credit at −0.67, which is an

Table 4.1: Descriptive statistics for the shock and the canonical response variables

Variable	*N*	Mean	SD	Min	Max
Shock and response variables					
ECB easing surprise	249	0.046	2.414	−21.122	13.661
house-purchase lending growth	250	3.620	3.173	−0.559	12.174
pure new house-purchase loans	99	10.920	0.189	10.518	11.310
mortgage lending spread	250	2.249	0.906	−0.240	3.752
NFC lending spread	250	2.104	0.451	1.020	3.125
NFC credit	250	15.328	0.110	14.977	15.480
real DAX	250	8.997	0.329	8.185	9.653
ECB assets	250	14.941	0.667	13.704	15.994
real wage tracker excl. one-offs	154	0.118	2.041	−6.926	2.764
German industry real wage bill growth	238	0.408	3.295	−13.373	7.112
employment expectations	250	100.135	10.012	50.800	116.200
Control variables					
EA20 annual HICP inflation	238	2.099	1.968	−0.619	10.102
Deposit facility rate, monthly average	250	0.773	1.415	−0.500	4.000

Notes: Monthly observations, January 2005 through October 2025. Units follow the variable definitions in this chapter. Sample sizes differ because the pure-new-loan, wage-tracker, and wage-bill series start later than the panel. The local projections also control for two lags of the response variable itself. Source: own calculations.

early sign of the sectoral differentiation the results chapter documents. These are unconditional correlations in levels and growth rates, so they describe co-movement in the panel rather than responses to policy news. The local projections are what separate the two.

Table 4.2: Pairwise correlations among the shock and the canonical response variables

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) ECB easing surprise										
(2) house-purchase lending growth	0.06									
(3) pure new house-purchase loans	0.07	0.69*								
(4) mortgage lending spread	0.08	0.06	0.53*							
(5) NFC lending spread	0.10	0.12*	0.51*	0.91*						
(6) NFC credit	−0.11	−0.67*	−0.06	−0.36*	−0.39*					
(7) real DAX	−0.06	−0.29*	0.10	−0.72*	−0.76*	0.46*				
(8) ECB assets	−0.14*	−0.43*	0.23*	−0.61*	−0.70*	0.71*	0.82*			
(9) real wage tracker excl. one-offs	0.12	−0.60*	−0.18	−0.02	0.00	−0.29*	0.03	−0.50*		
(10) German industry real wage bill growth	0.13*	−0.03	−0.15	0.14*	0.25*	−0.33*	−0.02	−0.26*	0.36*	
(11) employment expectations	0.02	0.35*	0.04	−0.36*	−0.32*	−0.16*	0.47*	0.23*	−0.43*	0.39*

Notes: Pearson correlations on pairwise complete monthly observations. * $p < 0.05$. Sample sizes vary across pairs because several series start later than the panel. Source: own calculations.

With the data architecture, the proxy choices, the sample, and the descriptive picture fixed, the next chapter turns to the response evidence.

5. Results

This chapter reports the estimated responses to a one-standard-deviation ECB target-factor easing surprise. The order follows the transmission chain of Equation (3.1). It moves from identification to housing credit, then to lending conditions and intermediation. Equity and asset markets follow, then the compensation-linked block, and the chapter closes with the synthesis and the hypothesis assessment.

5.1 Identification Validity

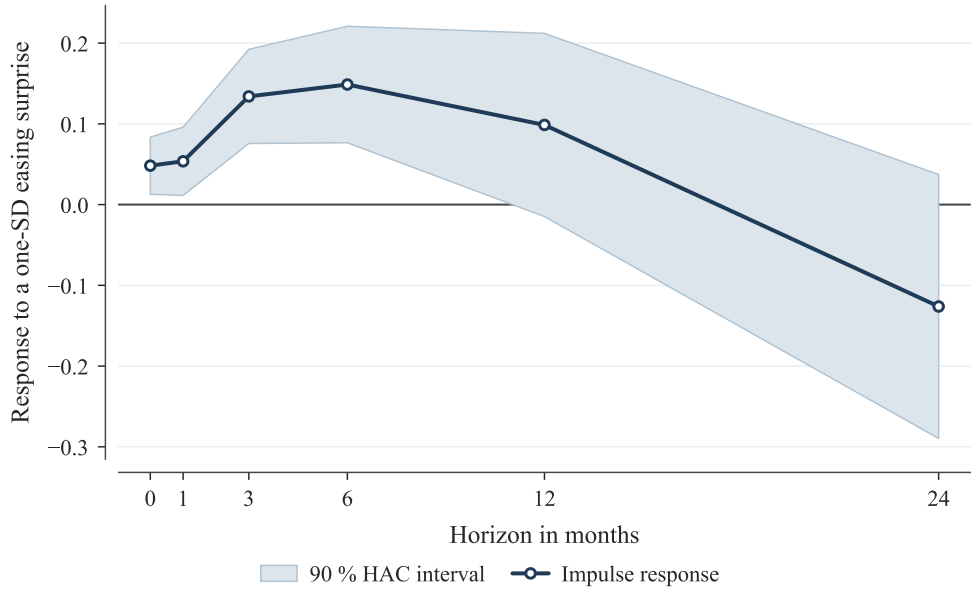
Chapter 3 builds the easing-signed target surprise and tests it. Two limits from that work set the frame for everything below. The surprise has real full-sample relevance for the policy stance, but the relevance is carried by a handful of large events and is not steady enough to treat the surprise as an instrument for the deposit rate or the balance sheet. The estimates are therefore responses to policy news, not to a cleanly isolated structural shock. The second limit is the information effect. An ECB announcement can move markets both through the accommodation it delivers and through what it reveals about the economy. The relevance diagnostics (Table 6.2) and the event-level screen (Table 6.1), both developed in Chapter 6, carry these boundaries into the results. The screen returns in the equity block, where it helps to interpret a negative equity response to easing.

The blocks below take the response paths one variable at a time, in the order of the transmission chain. House-purchase lending growth rises, accumulates through the first year, and gives back part of that gain in the second (Figure 5.1). Mortgage and corporate lending spreads widen and hold (Figures 5.2 and 5.3). The wage tracker is weaker and loses precision at the long horizon (Figure 5.7). Employment expectations rise on impact and reverse by two years (Figure 5.8). The real DAX moves the other way (Figure 5.6).

5.2 Housing-Credit Transmission

Housing credit is the core result, and house-purchase lending growth is the single clearest response in the thesis.

Figure 5.1: Baseline impulse response of house-purchase lending growth to a one-standard-deviation ECB easing surprise.



Source: own calculations.

The response is positive from impact through one year and peaks at the six-month horizon near 0.15, where the 90 and 95 percent intervals both exclude zero ($p = 0.001$). The single-horizon coefficient then fades and turns negative (about −0.13) at two years. The accumulated path carries the more important message. It builds to about 0.48 by twelve months and then falls back to about 0.36 at twenty-four months, with 90 and 95 percent HAC intervals that stay clear of zero at both horizons. The fall is a partial reversal. About a quarter of the first-year accumulation is given back in the second year, which is what the negative single-horizon coefficient at two years implies. The dependence-aware bootstraps in Table 6.4 agree with the HAC intervals through twelve months and part ways at two years, where both the moving-block and the wild interval include zero. The two-year accumulated response is therefore positive under every method but precisely estimated only under the HAC approximation. The response of house-purchase lending growth to an easing surprise is positive for the better part of a year, partly reverses in the second year, and its accumulated response stays positive at two years. The

claim of durability is firmest through twelve months. That is medium-run credit accommodation, read from the housing-credit channel, not from prices.

Sign stability is the share of the 42 cells formed by the seven event and exclusion samples of Section 6.3 and the six horizons in which the coefficient carries its modal sign. For house-purchase lending growth it is 0.86, and the accumulated direction at twelve and twenty-four months is the same in all seven samples, so the positive accumulated path is not the product of one unusual sample.

The same channel can look different in a second series, pure-new house-purchase loans, which measure monthly originations rather than the growth of the outstanding book.

The pure-new series shows no response that is precise under the HAC intervals and cannot carry a persistence claim. The point responses are small and change sign across horizons. The two-year point of about -0.03 has an HAC interval that includes zero, and every accumulated interval includes zero. The bootstrap band is tighter and excludes zero at the two-year point, which is why the series is kept as supporting evidence rather than dropped.

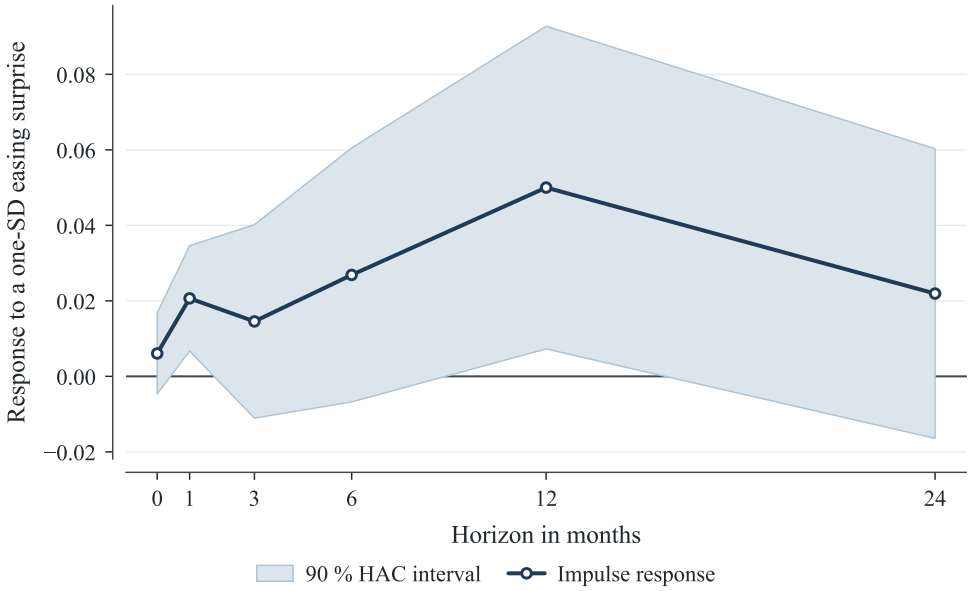
5.3 Lending Conditions and Intermediation

The intermediation block asks whether lending conditions move in a way that makes the housing-credit response believable. The lending spreads give the cleanest answer.

The mortgage lending spread over the deposit rate is positive at every reported horizon and peaks near 0.05 at one year, where its interval clears the 90 percent bar. Its accumulated response is positive and excludes zero at both 90 and 95 percent at six, twelve, and twenty-four months under the HAC intervals and under the moving-block bootstrap, though not under the wild bootstrap. Sign stability is a perfect 1.00 and the peak horizon does not move across the sample. This is the most stable response in the whole set. The economics is direct. After an easing surprise the margin banks earn over the policy floor widens, while the mortgage rate itself shows no precise response. A wider margin is the price-side counterpart to the housing-credit quantity response in the previous block. Both move in the same direction after the same easing surprise, which is what the intermediation reading expects.

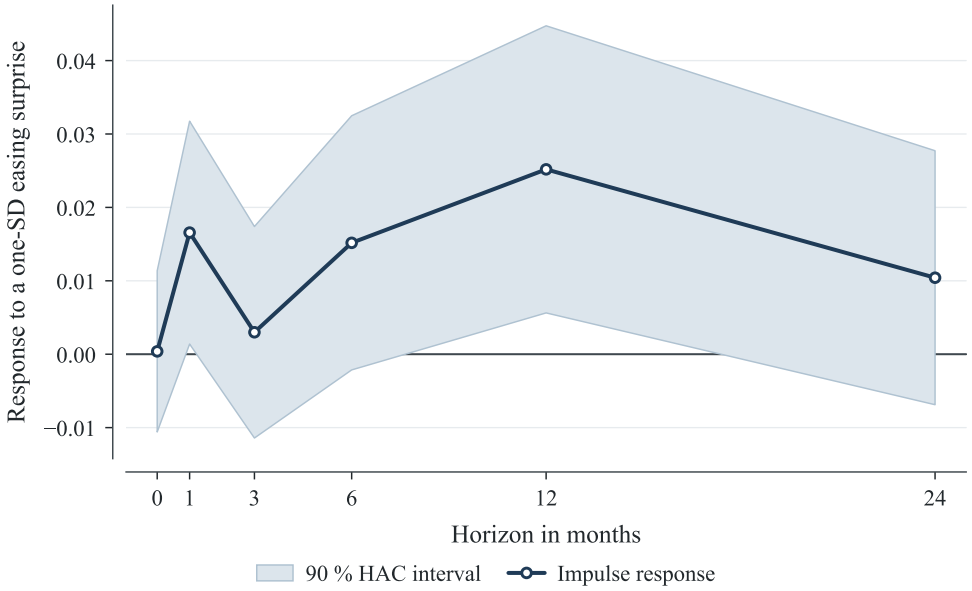
The NFC lending spread follows the same shape at about half the magnitude. It peaks near 0.025 at one year with an interval that excludes zero, and its accumulated path clears both 90 and 95 percent through two years under the HAC intervals, while the moving-block bootstrap

Figure 5.2: Baseline impulse response of the mortgage lending spread to a one-standard-deviation ECB easing surprise.



Source: own calculations.

Figure 5.3: Baseline impulse response of the NFC lending spread to a one-standard-deviation ECB easing surprise.

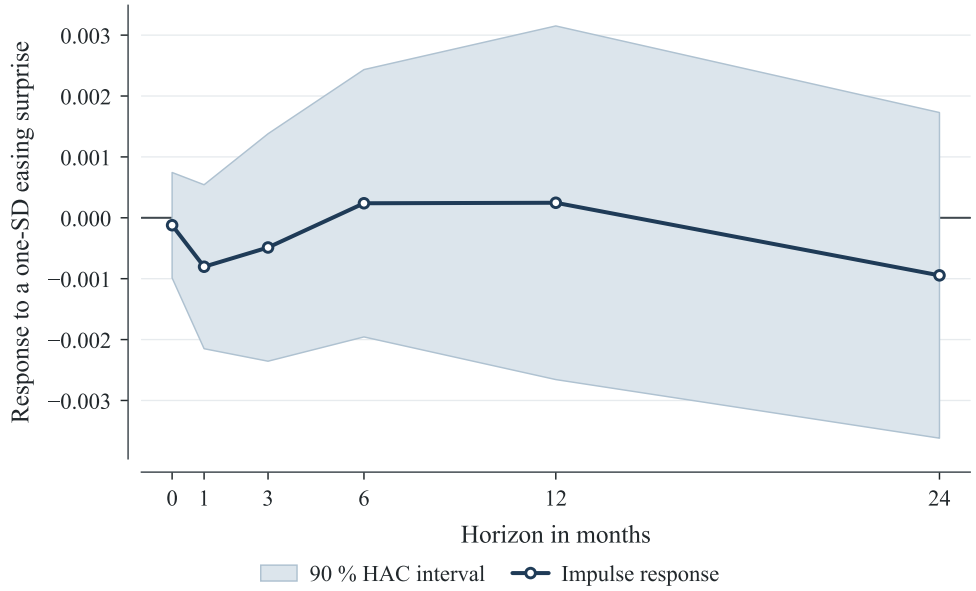


Source: own calculations.

confirms only the twelve-month exclusion. Corporate-side margins adjust in the same direction as mortgage margins, durably but by less. Lending-rate levels are estimated less precisely, so the result rests on the margin over the deposit rate rather than on a mechanical move in the rate level.

Broad credit quantities are weaker and sit to one side of the mechanism.

Figure 5.4: Baseline impulse response of Euro Area NFC loan stocks to a one-standard-deviation ECB easing surprise.



Source: own calculations.

Euro-area NFC and household credit stocks have small response paths and accumulated intervals that include zero at every horizon. Easing still reaches credit, but transmission is sectorally differentiated. Housing-credit growth and lending margins respond and persist, while broad productive-credit quantities do not move with any precision on this shock. The NFC-credit path is too imprecise to support a direction, with sign stability near 0.52 and a cumulative-direction score of 0.50.

The sequential timing outputs add a second view of the intermediation block. They are ordered associations, not evidence of structural mediation.

Table 5.1 reports the eight pathways. In each pathway the upstream variable is taken one or two months after the shock and the downstream variable three or six months after it, so the order is set by the design. The estimate asks whether the early upstream move carries information about the later downstream move once the shock is held fixed. Three quantity links are positive

Table 5.1: Ordered timing associations along the intermediation chain

Upstream	Downstream	h_u	h_d	λ	p
mortgage lending spread	house-purchase lending growth	1	6	−1.470	0.088
NFC lending spread	NFC credit	1	6	0.028	0.077
ECB assets	NFC credit	1	3	0.054	0.002
household credit	house-purchase lending growth	1	3	98.465	0.000
house-purchase lending growth	pure new house-purchase loans	1	3	0.152	0.000
NFC credit	real wage tracker excl. one-offs	2	6	2.331	0.855
employment expectations	real wage tracker excl. one-offs	1	6	−0.037	0.042
real DAX	real wage tracker excl. one-offs	1	6	−2.588	0.073

Notes: λ is the association between an earlier upstream movement and a later downstream movement, holding the policy-news shock fixed. This is ordered timing evidence and not structural mediation. Source: own calculations.

and precise: household credit to house-purchase lending growth, house-purchase lending growth to new mortgage flows, and ECB assets to NFC credit. The price-to-quantity links are weaker. The NFC spread to NFC credit link is positive at the ten percent level. The mortgage-spread to house-purchase growth link is negative and marginal, at -1.47 with $p = 0.088$, so months in which the mortgage margin widened more in the first month saw somewhat weaker house-purchase growth six months on. That is the sign a price-quantity relation should carry, and it is not evidence that the margin drives the credit expansion. The three links into wage pressure are negative or imprecise, which matches the weak and late compensation response reported below.

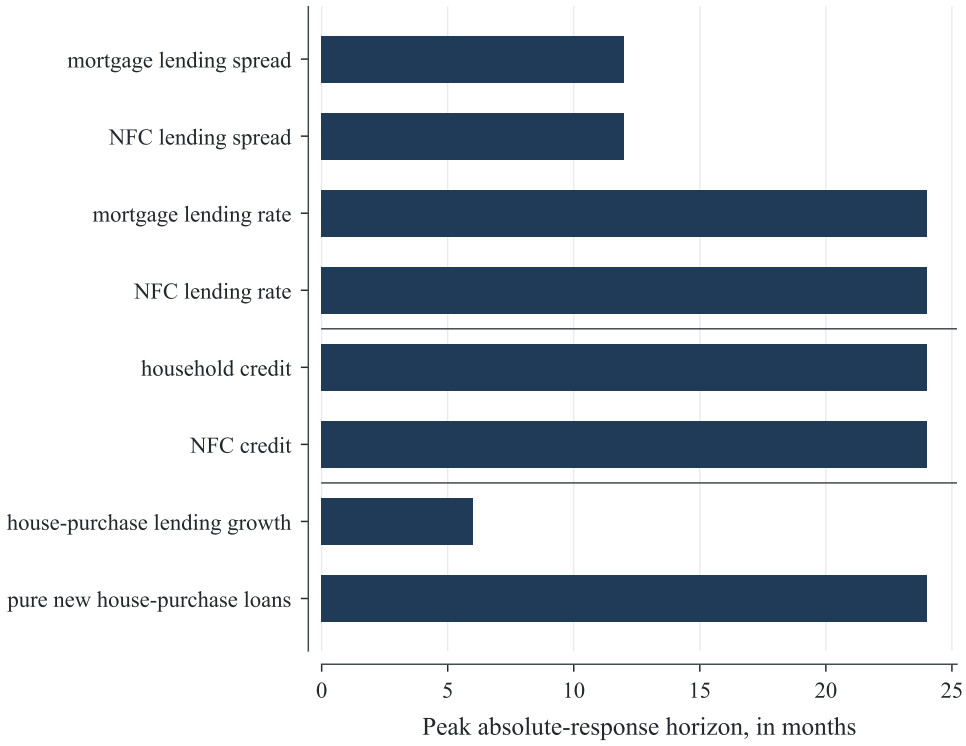
The peak horizons in Figure 5.5 complete the picture. House-purchase lending growth peaks at six months, the two spreads at twelve, and the credit stocks and lending rates only at twenty-four. Both spreads respond within the first month, as housing credit does, but their response builds more slowly and peaks later. What the intermediation block documents is that the price side and the quantity side respond to the same shock, in the same direction, over the same year. It does not show one leading the other, and monthly aggregates could not separate one bank's decision from one household's response even if it did.

5.4 Equity and Asset-Market Responses

The financial-market block does not support the simple idea that easing lifts every asset price.

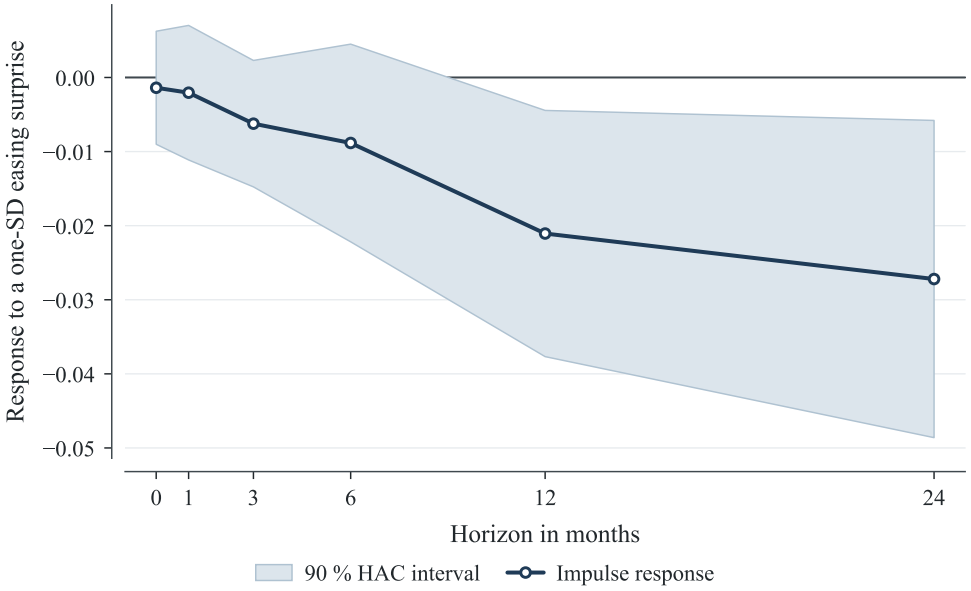
The real DAX is small and insignificant on impact, then increasingly negative from three months out. The twelve- and twenty-four-month coefficients, near -0.021 and -0.027, exclude zero at 90 and 95 percent. The accumulated response excludes zero from one year, reaching about -0.067 at two years, and the two-year exclusion holds under the moving-block and the

Figure 5.5: Peak timing evidence for banking, credit, and housing-finance response variables.



Source: own calculations.

Figure 5.6: Baseline impulse response of the real DAX to a one-standard-deviation ECB easing surprise.



Source: own calculations.

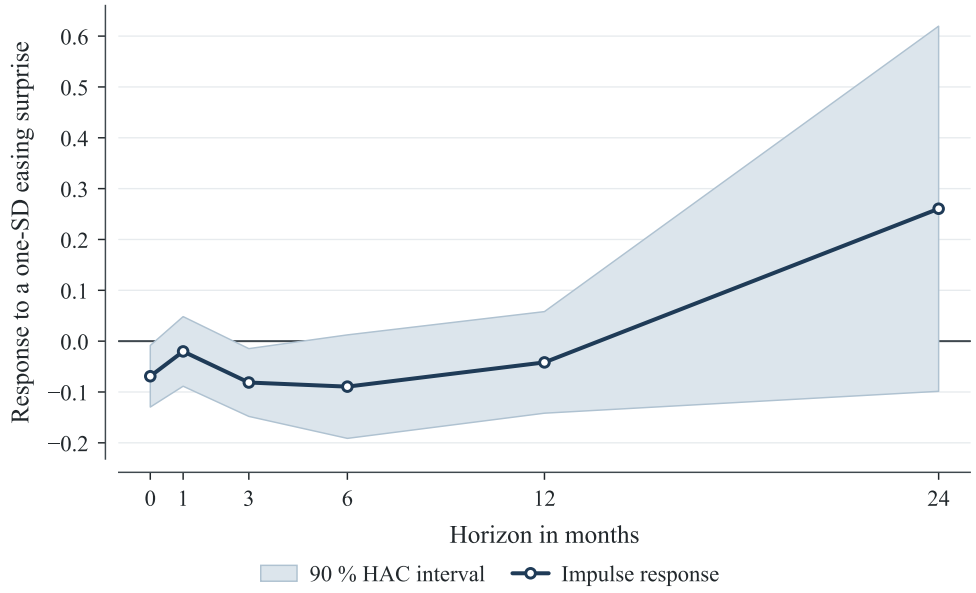
wild bootstrap as well. Sign stability is 0.90 and the cumulative direction is perfectly consistent. This is the sharpest sign puzzle in the results. Real equity prices fall after an easing surprise. The information screen is the natural reading, and the response is treated as an information-effect boundary here. The deeper mechanism is left to the discussion.

ECB assets are negative through two years as well, but the intervals include zero at every horizon. They give liquidity-stock context rather than a firm response. Taken together, the asset-market variables mark a boundary. Some financial prices respond, but the durable positive mechanism sits in the housing-credit and lending-condition blocks, not in equities or the balance-sheet stock.

5.5 Compensation-Linked and Real-Economy Responses

The compensation-linked block is the real-economy comparison. The euro-area wage tracker and employment expectations are weaker and less consistent than the financial blocks above. The German industry wage bill is not.

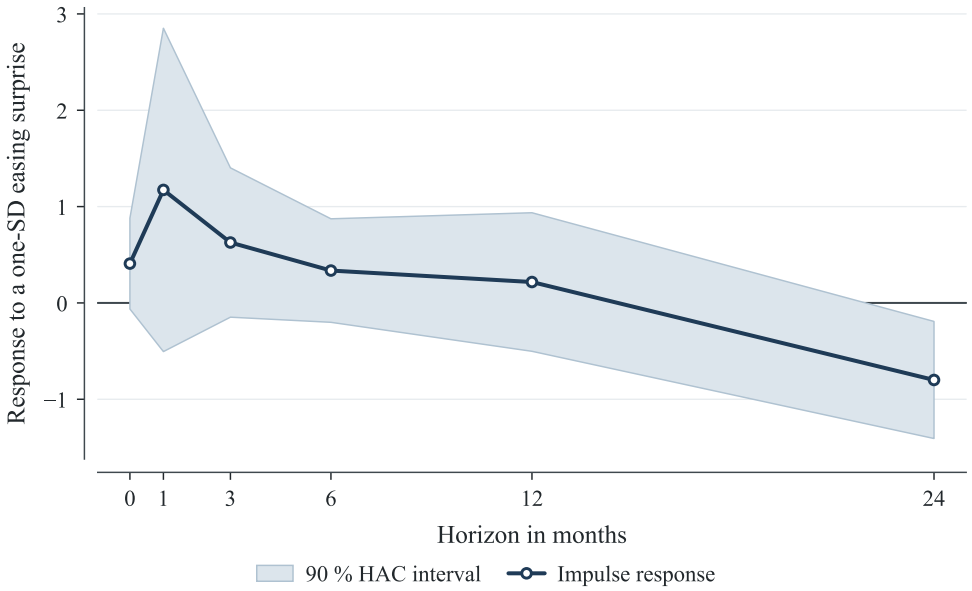
Figure 5.7: Baseline impulse response of the real ECB Wage Tracker excluding one-offs to a one-standard-deviation ECB easing surprise.



Source: own calculations.

The real ECB Wage Tracker excluding one-offs is negative from impact through one year and turns positive only at the two-year point, where the HAC interval includes zero even though the

Figure 5.8: Baseline impulse response of employment expectations to a one-standard-deviation ECB easing surprise.



Source: own calculations.

bootstrap band does not. Its accumulated response is negative and excludes zero at six and twelve months under the HAC intervals and the wild bootstrap, near −0.26 and −0.30, but not under the moving-block bootstrap, and it returns toward zero and includes zero at two years under every method. Real compensation growth dips after easing and the dip fades. The short-horizon fall is consistent with prices moving faster than nominal pay right after an easing surprise, but the long-horizon estimate is too imprecise, on a 152-observation sample, to claim durable income transmission. Sign stability is only 0.67.

Employment expectations move earlier and by more. They are positive from impact through one year, with an impact reading near 0.41 and a one-month peak near 1.17, then reverse to a negative and HAC-significant value at two years. The accumulated path is strongly positive and excludes zero at six and twelve months under the HAC intervals and the moving-block bootstrap, though not under the wild bootstrap, and includes zero at two years under every method. Employment expectations rise quickly after an easing surprise and then give the improvement back. This is short-horizon labor sentiment, not durable transmission into employment.

The German industry real wage bill does not fade. It rises on impact and peaks at one month, and its accumulated response keeps its sign and excludes zero at every horizon through two years, near 1.49 at h24. That holds under the HAC interval and under both bootstraps from one

month on (Tables 5.4 and 6.4). It is sectoral and national rather than euro-area wide, but it is a compensation-linked response, and its response is durable.

5.6 Persistence Synthesis

Sign, accumulated response, precision, and stability line up most fully for four responses. Table 5.2 sets the variables side by side.

Table 5.2: Main synthesis table for persistence evidence

Variable	Sign	Peak horizon	Cumulative persistence	Stability	Interpretation
House-purchase lending growth	Positive	h6	Positive h24 after a partial reversal from h12, 90% HAC CI excludes zero, bootstrap through h12	High: sign 0.86, cumulative 1.00	Core housing-credit response and main empirical contribution.
Mortgage spreads	Positive	h12	Positive h24, 90% CI excludes zero (HAC and moving block)	Very high: sign 1.00, cumulative 1.00	Stable lending-condition margin supporting intermediation.
NFC spreads	Positive	h12	Positive h24, 90% HAC CI excludes zero, bootstrap at h12 only	High: sign 0.90, cumulative 1.00	Smaller but durable corporate credit-price adjustment.
NFC credit	Mixed	h24	Negative h24, CI crosses zero	Weak: sign 0.52, cumulative 0.50	Productive-credit quantity response too imprecise for a persistence claim.
DAX	Negative	h24	Negative h24, 90% CI excludes zero (HAC and bootstrap)	High: sign 0.90, cumulative 1.00	Information-sensitive market response, not asset-appreciation evidence.
ECB assets	Negative	h24	Negative h24, CI crosses zero	Moderate: sign 0.79, cumulative 0.86	Liquidity-stock context, not a strong persistence pillar.
Wage tracker	Mixed	h24	Negative h24, CI crosses zero	Moderate: sign 0.67, cumulative 0.86	Euro-area compensation response attenuates by h24.
German wage bill	Positive	h1	Positive h24, 90% CI excludes zero (HAC and both bootstraps)	High: sign 0.81, cumulative 1.00	Sectoral compensation response, durable through h24.
Employment expectations	Positive then reverse	h1	Positive h24, CI crosses zero	Moderate: sign 0.76, cumulative 0.79	Short-horizon labor expectations, not durable income transmission.

Notes: Peak horizons, accumulated responses, and stability shares summarize the estimates reported in Tables 5.3 and 5.4. CI refers to the HAC interval under the independence approximation, bootstrap to the joint moving-block and wild intervals in Table 6.4. Source: own calculations.

Read down the table and three groups appear. House-purchase lending growth, the two lending spreads, and the German wage bill form the durable positive block, with accumulated HAC intervals that exclude zero through two years and high to perfect stability. The two spreads keep accumulating through the second year. Housing credit does not. Its accumulated response peaks at twelve months and partly reverses, and in shape alone its path resembles that of employment expectations, which also peak at twelve months and give back a comparable share of the gain by two years. What separates the two is precision and stability, not shape. The housing-credit

accumulation keeps its sign in every sample and horizon and its two-year HAC interval excludes zero, while employment expectations lose both. The DAX is just as durable but in the opposite direction, which places it on the information-effect boundary rather than in the appreciation story. NFC credit, ECB assets, the wage tracker, and employment expectations mark the limits of the evidence, each losing either precision, sign stability, or long-horizon exclusion. The exclusions in the table are HAC exclusions. Under the dependence-aware bootstraps in Table 6.4 the two-year exclusion survives for the mortgage spread, the German wage bill, and the DAX, while the housing-credit exclusion survives through twelve months.

The underlying estimates are reported in full in Table 5.3, which gives the horizon-level coefficients and HAC standard errors for every response, and in Table 5.4, which gives the accumulated paths and marks where the 90 percent interval excludes zero. Reading the two together shows the pattern the synthesis compresses. Housing credit, the lending spreads, and the German wage bill hold their accumulated HAC exclusion at the long horizons, while the wage tracker, employment expectations, and broad credit lose it.

5.7 Hypothesis Assessment

Table 5.5 sets out the three assessments. H1 is supported at short and medium horizons in the full-sample estimates. House-purchase lending growth responds on impact and peaks at six months, and its accumulated response excludes zero from impact through twelve months under the HAC interval and both bootstraps. H2 is supported for the canonical wage tracker, with a proxy boundary. The German industry wage bill, kept as a sectoral robustness proxy, keeps its sign and excludes zero from one month through two years under every interval method, so the verdict depends on which compensation proxy is treated as canonical. The conditional word for H3 is deliberate. The claim is not that every coefficient is invariant, only that the ordering of durable responses survives most clearly for house-purchase lending growth and the lending spreads.

Table 5.3: Baseline monthly impulse responses to a one-standard-deviation ECB easing surprise

	Horizon in months					
Response	0	1	3	6	12	24
house-purchase lending growth	0.048**	0.054**	0.134***	0.149***	0.099	−0.126
	(0.022)	(0.026)	(0.035)	(0.044)	(0.069)	(0.099)
pure new house-purchase loans	−0.007	0.013	0.001	0.020	−0.001	−0.031
	(0.012)	(0.011)	(0.013)	(0.013)	(0.013)	(0.027)
mortgage lending spread	0.006	0.021**	0.015	0.027	0.050*	0.022
	(0.007)	(0.008)	(0.016)	(0.020)	(0.026)	(0.023)
NFC lending spread	0.000	0.017*	0.003	0.015	0.025**	0.010
	(0.007)	(0.009)	(0.009)	(0.011)	(0.012)	(0.011)
NFC credit	0.000	−0.001	0.000	0.000	0.000	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
real DAX	−0.001	−0.002	−0.006	−0.009	−0.021**	−0.027**
	(0.005)	(0.006)	(0.005)	(0.008)	(0.010)	(0.013)
ECB assets	−0.004	−0.002	−0.003	−0.003	−0.010	−0.011
	(0.003)	(0.004)	(0.006)	(0.007)	(0.012)	(0.015)
real wage tracker excl. one-offs	−0.069*	−0.020	−0.081**	−0.089	−0.042	0.260
	(0.037)	(0.042)	(0.041)	(0.062)	(0.061)	(0.218)
German industry real wage bill growth	0.205**	0.422	0.347*	0.403***	0.216	−0.105
	(0.096)	(0.270)	(0.191)	(0.118)	(0.175)	(0.117)
employment expectations	0.408	1.173	0.627	0.336	0.217	−0.800**
	(0.287)	(1.020)	(0.471)	(0.327)	(0.437)	(0.369)

Notes: Local-projection coefficients scaled to a one-standard-deviation easing surprise. HAC (Newey–West) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own calculations.

Table 5.4: Accumulated responses at the medium and long horizons

	Horizon in months		
Response	6	12	24
house-purchase lending growth	0.385†	0.483†	0.357†
pure new house-purchase loans	0.027	0.025	−0.006
mortgage lending spread	0.068†	0.118†	0.140†
NFC lending spread	0.035†	0.060†	0.071†
NFC credit	−0.001	−0.001	−0.002
real DAX	−0.019	−0.040†	−0.067†
ECB assets	−0.012	−0.022	−0.032
real wage tracker excl. one-offs	−0.260†	−0.302†	−0.041
German industry real wage bill growth	1.377†	1.593†	1.488†
employment expectations	2.545†	2.762†	1.962

Notes: Running sum of the normalized impulse responses through each horizon. Intervals combine the horizon-level HAC standard errors under an independence approximation across horizons. † marks accumulated responses whose 90 percent HAC interval excludes zero. Table 6.4 reports the dependence-aware bootstrap intervals for the same accumulated responses. Source: own calculations.

Table 5.5: Empirical assessment of the three thesis hypotheses

Hypothesis	Evidence	Assessment
H1: Housing-credit persistence	House-purchase lending growth is positive on impact and peaks at h6. Its accumulated response excludes zero from h0 through h12 under the HAC interval and both bootstraps.	Supported at short and medium horizons.
H2: Compensation-linked attenuation	The canonical wage tracker moves at short and medium horizons but loses h24 cumulative exclusion, while employment expectations reverse. The German industry wage bill, a sectoral robustness proxy, keeps its sign and excludes zero from h1 through h24 under every interval method, which is the proxy boundary on this verdict.	Supported with proxy boundary.
H3: Differential financial-intermediation transmission	Housing-credit growth and lending spreads are stronger and more persistent than NFC credit quantities and the euro-area wage tracker.	Conditionally supported.

Notes: H1 to H3 are Hypotheses 1 to 3 of Chapter 1, derived in Section 3.4. CI refers to the HAC interval under the independence approximation, bootstrap to the joint moving-block and wild intervals in Table 6.4. Source: own assessment based on own calculations.

5.8 Regime Comparisons

The regime split is descriptive. It shows how the same target surprise maps into different observed conditions across policy eras, not a separate identified experiment. The four eras are pre-QE through January 2014, QE from February 2014 to December 2019, COVID from January 2020 to December 2021, and tightening from January 2022 to the end of the sample.

Pre-QE estimates carry strong positive accumulated house-purchase lending growth through two years. Under QE the same surprise lands in different conditions, including negative house-purchase lending growth and negative NFC spreads. COVID is visible only at short horizons. The tightening era shows positive housing-credit and spread accumulation alongside negative wage-pressure accumulation. These patterns speak to historical dependence. They do not replace the full-sample estimates, and the robustness chapter returns to them under formal event filters.

The responses are now in place. The next chapter asks whether they survive changes to the specification, the shock, and the sample.

6. Robustness & Diagnostics

The results chapter ranked the channels by how clearly and durably they respond to an ECB easing surprise. This chapter asks whether that ranking survives when the specification, the shock, and the sample are changed. Here robustness has the same meaning it has in the research question. It is durability of the sign and of the accumulated direction, not identical coefficients in every cut of the data. A response fails the test when either one flips. The point of the exercise is to see whether the comparative ordering in Table 5.2 is a feature of the data or an artifact of one specification.

Four responses cluster at the stable end of the checks below. Mortgage spreads, NFC spreads, house-purchase lending growth, and the real DAX hold their direction across the widest set of checks. These are the intermediation and housing-credit variables that carry the mechanism, plus the equity price that sits alongside it. The compensation-linked proxies stay informative but move more with the specification.

6.1 Alternative Lag Structures

The pipeline does not run an exhaustive lag grid, and this is a design choice rather than an omission. The horizons are the irregular set $\{0, 1, 3, 6, 12, 24\}$, so the natural threat is not a misspecified lag order but the overlap between neighbouring projections. That overlap is handled by the HAC and bootstrap machinery set out in Chapter 3, together with the rolling and recursive windows re-estimated below.

Two layers carry that stress. The seven event and exclusion samples defined in Section 6.3 re-estimate every canonical response at every horizon, and rolling and recursive windows re-estimate four of them. The rolling windows are 84 months long and move one month at a time. The recursive estimates start from the first 60 months and extend in six-month steps. Both run for house-purchase lending growth, pure-new loans, the wage tracker, and the DAX at the six- and twelve-month horizons. Across the sample cuts the same four responses stay put. House-purchase lending growth, mortgage spreads, NFC spreads, and the DAX keep their modal sign in at least 86 percent of the sample-by-horizon cells and their accumulated direction in every

sample. NFC credit and the wage tracker do not, and their signs move once the event sample or the exclusion window shifts. The rolling windows are the harsher test. House-purchase lending growth keeps its positive sign in about three quarters of the 84-month windows at six months and two thirds at twelve, and in every recursive sample at six months. The wage tracker is negative in almost every six-month window. The DAX sign at six months is close to a coin toss across windows, while at twelve months it is negative in two thirds of them, which is why the equity result is read as a medium- and long-horizon result.

The channels that answer the research question most clearly are also the ones whose direction is least sensitive to how the dynamics are specified.

6.2 Alternative Shock Measures

The baseline shock is the easing-signed target surprise. The timing, forward-guidance, QE, and weighted-composite factors are re-estimated as alternative shocks to see how far the ordering depends on which piece of the surprise is used. The weighted composite sits closest to baseline. The DAX stays negative, house-purchase lending growth stays positive at h6 and h12, and the mortgage spread accumulates positive. The NFC spread is the one place the weighted composite departs from baseline, with an accumulated path that stays slightly negative even though its marginal response turns positive at the longer horizons. The forward-guidance, timing, and QE components change several signs, most visibly for wage pressure and for some housing-credit specifications. Component choice is material, which is the expected reading once the surprise is decomposed rather than treated as one object (Jarociński & Karadi, 2020). These alternative factors serve as diagnostics on the shock and leave the baseline in place. The baseline and alternative-shock estimates share the normalized IRF output reported in Table 5.3.

A separate check compares the surprise-based local projection against a plain OLS regression on the raw policy-rate change. The raw-rate regression is estimated for the ten canonical responses, which gives 60 comparable response-horizon cells. Read in the same direction, with a rate cut counted as easing, the two designs agree in 36 of those cells and disagree in 24. Full agreement is not expected. The surprise design uses easing-signed announcement news, while the raw policy-rate regression uses the realised rate move, so the two objects should not track each other coefficient by coefficient. Every cell is labelled as a qualitative direction check, and the file treats it as a sign comparison with no causal reading. The file compares raw signs, so

its match column records the opposite of the direction-adjusted reading used here. The lending spreads and the wage tracker agree at every horizon, and the DAX through twelve months. House-purchase lending growth agrees at six and twelve months, while employment expectations and the German wage bill disagree at nearly every horizon.

6.3 Sub-Sample Analysis

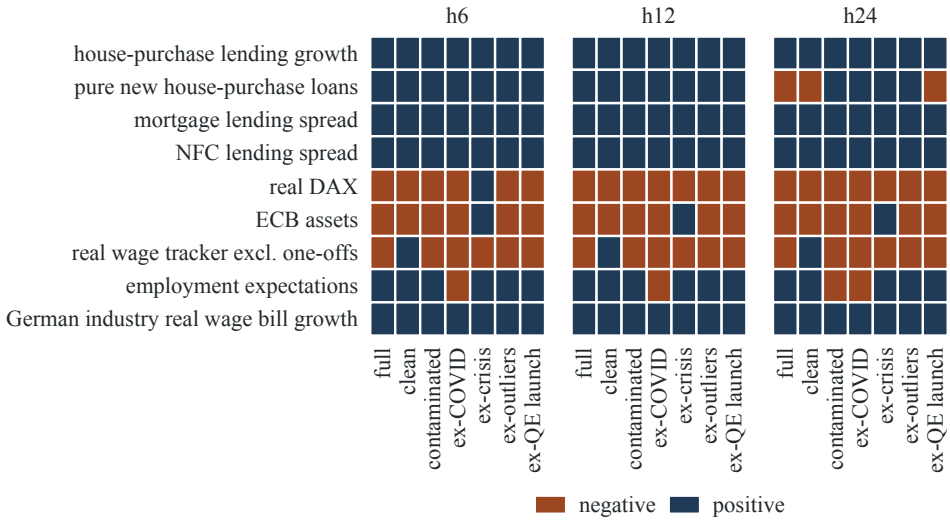
The sample is cut seven ways: the full sample, clean events only and contaminated events only, which rebuild the monthly shock from the two event classes of the screen, and four historical exclusions that drop the COVID era from January 2020 to December 2021, the crisis windows from September 2008 to June 2009, July 2011 to December 2012, and March 2020 to December 2021, the five months with the largest absolute surprises, and the QE launch from January to June 2015. The cumulative results are stable where it counts. House-purchase lending growth accumulates positive at h6, h12, and h24 in all seven samples. Mortgage and NFC spreads accumulate positive at the same three horizons in all seven. The DAX accumulates negative at h12 and h24 in all seven.

The fragile responses behave differently under the same cuts. The wage tracker changes accumulated sign between samples. It is positive in the clean-event sample and negative in the full and contaminated samples, so its long-run direction is not a fixed feature of the data. NFC credit is inconsistent across the seven cuts and never settles on one direction. These are the same two responses that moved under the lag and window checks.

One distinction matters for how the house-purchase result is read. The durable positive claim is about the accumulated path, not the marginal coefficient at every horizon. The marginal response for house-purchase lending growth is positive through h12 and turns negative at h24 in six of the seven filters, with only the crisis-window exclusion keeping it positive. The accumulated path still stays positive because the earlier horizons carry more weight. This is the same object seen in the baseline, where the h24 point coefficient is negative while the running sum remains positive. The directional matrix records both the marginal and the cumulative signs, which is why the two are kept separate.

Figure 6.1 shows why the claim is stated in accumulated terms. Housing-credit growth and the lending spreads keep their cumulative direction across the main filters at medium and long horizons. The wage tracker and employment expectations move with sample composition.

Figure 6.1: Robustness consistency matrix for cumulative response directions across event filters and historical exclusions.



Source: own calculations.

6.4 Identification Fragility

Not every announcement is a clean monetary signal, and the event screen quantifies how much of the sample is exposed. Of 244 events, 86 look like possible pure monetary shocks (35.2 percent), 81 show a mixed sign conflict between the rate factor and the yield response (33.2 percent), 68 look like possible information shocks (27.9 percent), and 9 are ambiguous zero responses (3.7 percent). Roughly a third of the events read as clean surprises and the rest carry some contamination or conflict. The share shifts by regime. The financial-crisis window has the highest share of possible information shocks, close to half of its events. These counts describe the full 244-event archive. The estimation samples use the narrower monthly identification window from Chapter 3, so the clean-event count there is a subset of this archive total.

Table 6.1: Information-effect classification of ECB announcement events

Event classification	Events	Share (%)
possible pure monetary shock	86	35.2
mixed sign factor yield conflict	81	33.2
possible information shock	68	27.9
ambiguous zero response	9	3.7
Total	244	100.0

Notes: Events are classified by comparing the sign of the rate factor with the yield and equity responses in the announcement window. Counts cover the full event archive. Source: own calculations.

The screen sorts the responses the same way the other checks do. House-purchase lending growth and the lending spreads keep their qualitative direction across clean and contaminated events. The DAX is more negative in the contaminated events, which is the pattern expected if easing announcements also carry adverse macroeconomic news. Because the screen itself uses the sign of the equity move, part of this difference is built into the classification, so the information-effect reading remains an interpretation rather than a tested result. The wage tracker changes accumulated sign between the clean and contaminated samples. So the housing-credit and lending-margin ordering survives the contamination screen, while the equity and wage readings are the ones that need the information-effect caveat attached.

6.5 Relevance Sensitivity

The strongest first-stage bridge is the end-of-month DFR change on the easing-signed target surprise, with $F = 17.07$ and partial $R^2 = 0.066$. The full-sample first stage clears the weak-instrument bar on the classical statistic. The problem is where the relevance comes from. Drop the five largest absolute surprises, which make up 28.8 percent of the total absolute shock, and the first-stage F falls to 5.34, about 0.31 of its full-sample value. The rolling median F is 8.99, below the Stock-Yogo-style thresholds, even though the sign of the first-stage coefficient never flips across the rolling windows. Relevance is real but concentrated in a handful of large events.

Table 6.2: Monthly first-stage relevance of the strongest candidate policy bridges

Surprise factor	Policy target	F	Partial R^2	N	Screen
target factor	deposit facility rate, end of month	17.07	0.066	246	fragile candidate
timing factor	ECB assets, log monthly average	10.99	0.044	244	fragile candidate
fg factor	deposit facility rate, monthly average	9.43	0.038	244	fragile candidate
timing factor	ECB assets, log end-of-month stock	7.05	0.029	244	fragile candidate
qe factor	deposit facility rate, end of month	5.73	0.040	142	fragile candidate
target factor	ECB assets, log monthly average	4.42	0.018	246	reject weak relevance
target factor	household credit	3.02	0.012	246	reject weak relevance
qe factor	NFC credit	2.50	0.018	142	reject weak relevance

Notes: Eight strongest bridges from the monthly first-stage tournament, ranked by F-statistic. No candidate clears the strength and stability screen jointly, which is why the thesis reports reduced-form responses to policy news. Source: own calculations.

The tournament makes the same point across the full field. None of the 45 candidate bridges passes cleanly. The five strongest are flagged as fragile candidates and the remaining forty are

rejected for weak relevance. A full-sample first stage is informative, but it is not steady enough to support a strong claim about one treatment variable when the relevance is event-concentrated and regime-sensitive (Stock & Yogo, 2005). The thesis takes the conservative route. The surprise is not upgraded into an instrument for the DFR, the shadow rate, ECB assets, or the credit aggregates, and the estimand stays the reduced-form policy-news response. This narrows the treatment interpretation. It does not weaken the persistence comparison, which never depended on a structural first stage.

6.6 Sign Consistency

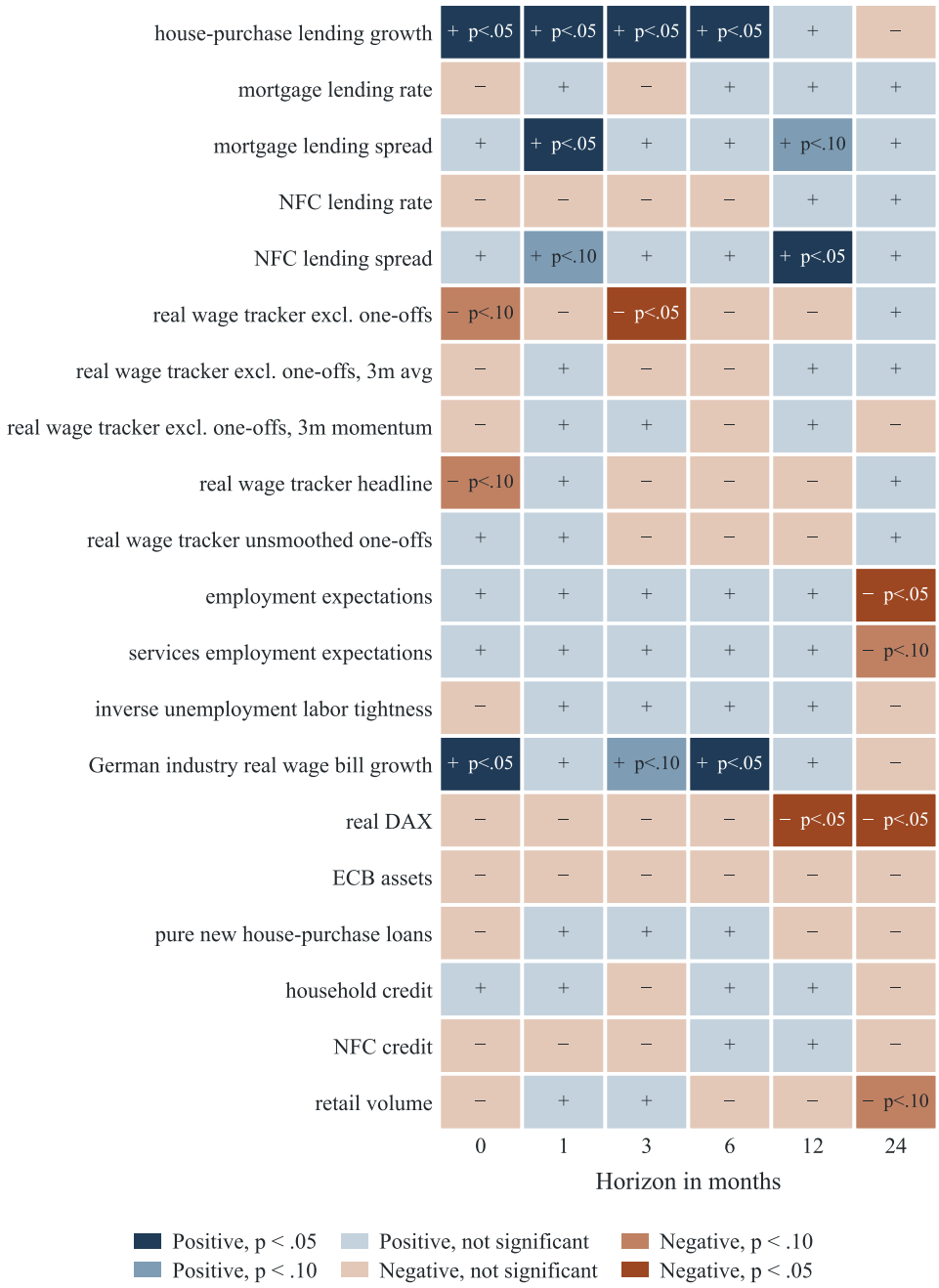
The sign-consistency ranking lines up with everything above and with the stability scores already tabulated in Table 5.2. The lending spreads and house-purchase lending growth sit at the top, the DAX matches them in strength but in the opposite direction, and NFC credit is the weakest, close to a coin toss and too imprecise to carry a persistence claim.

Individual coefficient significance is thin, and this is why the argument is built on sign patterns rather than single coefficients. Most horizon-by-response cells in the heatmap are not significant on their own. The clearest cases include house-purchase lending growth at the short horizons, the DAX at h12 and h24, the German wage bill at h0 and h6, employment expectations at h24, the mortgage spread at h1, and the NFC spread at h12. The persistence claim rests on the accumulated bands, where the cumulative confidence pattern is stronger. Under the HAC intervals, cumulative significance holds across the whole horizon path only for house-purchase lending growth and the German wage bill. It holds from h1 for the mortgage spread and from h6 for the NFC spread. For the DAX it holds only at h12 and h24. NFC credit, ECB assets, pure-new loans, household credit, and retail volume never reach it.

The long-horizon inference is joint rather than coefficient-by-coefficient because the projections overlap. Adjacent-horizon residual correlations run from 0.41 to 0.82 and reach 0.87 for NFC credit and 0.86 for house-purchase growth. That dependence is the reason the uncertainty layer uses moving-block and wild bootstraps and reads the cumulative bands as a set. The moving-block bootstrap uses 399 replications with a block length of 12, and the wild bootstrap gives a second view on the same responses.

Table 6.3 reports four tests on the residuals of every baseline regression, so that the choice between these interval methods rests on evidence rather than on convention. The first asks whether

Figure 6.2: Horizon-by-response significance heatmap for thesis-facing response variables.



Source: own calculations.

serial correlation remains beyond the HAC bandwidth, using the twelve autocorrelations at lags $h + 2$ to $h + 13$. It rejects at the five percent level in 45 of the 60 cells, and at every horizon for ECB assets, NFC credit, house-purchase lending growth, and pure new loans. The bandwidth $h + 1$ therefore covers the overlap that the design builds in, but not all of the dependence that remains, which is why the HAC intervals are reported alongside, and never instead of, the joint bootstraps, which resample all horizons together. The second and third tests concern the variance. Koenker's form of the Breusch–Pagan test rejects constant variance in 34 cells, and Engle's ARCH test with twelve lags rejects in 51, so the residual variance moves over time for almost every response. That is the condition the wild bootstrap is built for, and it is the reason the wild interval is reported next to the moving-block interval rather than treated as a secondary check. The fourth test, Jarque–Bera, rejects normality in 48 cells, with excess kurtosis the cause in 47 of them, which is why every bootstrap interval is a percentile interval and none relies on a normal approximation.

Table 6.3: Residual diagnostics for the baseline local projections

		Horizons (months) at which the test rejects at 5 percent			
Response	N	Serial corr.	Heterosk.	ARCH	Non-normal
house-purchase lending growth	246	all	1, 3, 6, 12, 24	all	0, 1, 3
pure new house-purchase loans	97	all	3, 6	12	none
mortgage lending spread	246	6, 12, 24	1, 3, 6, 12, 24	all	all
NFC lending spread	246	0, 12, 24	1	all	all
NFC credit	246	all	6, 12, 24	1, 3, 6, 12, 24	0, 1, 3, 6, 12
real DAX	246	6, 12, 24	all	1, 3, 6, 12, 24	all
ECB assets	246	all	0, 1, 3	all	0, 1, 3, 24
real wage tracker excl. one-offs	152	0, 1, 3, 6, 12	0, 1, 6, 12, 24	1, 3, 6, 12, 24	all
German industry real wage bill growth	235	0, 1, 3, 24	1, 3	all	all
employment expectations	246	3, 6, 12	0, 1	1, 3, 6, 12, 24	all

Notes: Baseline target-shock regressions, full sample. N is the estimation sample at $h = 0$. Serial corr.: Ljung–Box test on the twelve residual autocorrelations at lags $h + 2$ to $h + 13$, just beyond the Newey–West bandwidth $h + 1$. Heterosk.: Koenker's studentized Breusch–Pagan test of the squared residuals on the regressors. ARCH: Engle's LM test with twelve lags. Non-normal: Jarque–Bera. "all" means every one of the six horizons. Source: own calculations.

Read together, the diagnostics overturn no result, but they fix the order in which the intervals should be trusted at the long horizons. Where the HAC interval and the joint bootstraps agree, as they do for housing credit through twelve months, the conclusion is firm. Where they disagree, as they do for housing credit at twenty-four months, the bootstrap reading takes precedence, and Chapter 5 is written that way.

Table 6.4: Accumulated responses: 90 percent intervals from the dependence-aware bootstraps

		Horizon in months		
Response	Bootstrap	6	12	24
house-purchase lending growth	moving block	[0.167, 0.545]†	[0.109, 0.731]†	[-0.078, 0.734]
	wild	[0.143, 0.644]†	[0.151, 0.819]†	[-0.063, 0.809]
pure new house-purchase loans	moving block	[-0.042, 0.082]	[-0.065, 0.091]	[-0.088, 0.073]
	wild	[-0.059, 0.101]	[-0.068, 0.107]	[-0.085, 0.063]
mortgage lending spread	moving block	[0.016, 0.134]†	[0.039, 0.218]†	[0.034, 0.267]†
	wild	[-0.009, 0.148]	[-0.001, 0.234]	[-0.016, 0.293]
NFC lending spread	moving block	[-0.004, 0.081]	[0.008, 0.125]†	[-0.011, 0.155]
	wild	[-0.031, 0.106]	[-0.024, 0.149]	[-0.040, 0.176]
NFC credit	moving block	[-0.005, 0.003]	[-0.008, 0.006]	[-0.013, 0.009]
	wild	[-0.008, 0.006]	[-0.012, 0.010]	[-0.018, 0.013]
real DAX	moving block	[-0.044, 0.011]	[-0.075, 0.000]	[-0.111, -0.022]†
	wild	[-0.044, 0.010]	[-0.074, -0.006]†	[-0.123, -0.011]†
ECB assets	moving block	[-0.035, 0.015]	[-0.058, 0.021]	[-0.084, 0.026]
	wild	[-0.049, 0.028]	[-0.078, 0.040]	[-0.114, 0.058]
real wage tracker excl. one-offs	moving block	[-0.530, 0.061]	[-0.726, 0.175]	[-0.578, 0.533]
	wild	[-0.475, -0.014]†	[-0.572, -0.001]†	[-0.362, 0.332]
German industry real wage bill growth	moving block	[0.626, 2.136]†	[0.721, 2.489]†	[0.536, 2.458]†
	wild	[0.403, 2.356]†	[0.404, 2.730]†	[0.246, 2.691]†
employment expectations	moving block	[1.018, 4.632]†	[0.579, 5.403]†	[-0.462, 4.863]
	wild	[-0.365, 6.132]	[-0.810, 7.365]	[-1.582, 6.679]

Notes: Percentile intervals for the running sum of the normalized responses to the target easing surprise from 399 joint replications that resample the residuals of all six horizons together, in moving blocks of twelve months or with wild sign flips. † marks intervals that exclude zero. Compare with the HAC intervals under the independence approximation in Table 5.4. Source: own calculations.

Read together, these checks say the persistence claim rests on sign-consistent accumulated paths under HAC and bootstrap uncertainty, not on any one coefficient reaching significance in isolation (Jordà, 2005; Newey & West, 1987).

7. Discussion

Chapters 5 and 6 leave a clear pattern. House-purchase lending growth shows the most durable positive response to an ECB easing surprise, the two lending spreads move with it, the compensation-linked variables respond less consistently, and the real DAX falls. Tables 5.2 and 5.5 summarize that pattern. This chapter asks what it means. Each section takes one block of the evidence, reads it as economics, and sets it against what earlier studies found. The reading stays reduced-form and comparative throughout. The chapter closes with the limits of the design and the research agenda that follows from them.

7.1 Housing-Credit Transmission as the Core Result

The main result is that ECB policy-news surprises leave their clearest and most persistent trace in house-purchase lending growth. The response is positive on impact, peaks at six months, and stays positive through the first year. The single-horizon estimate turns negative at two years, so part of the first-year accumulation reverses, and the accumulated response falls from about 0.48 to about 0.36 while staying positive. Its HAC interval excludes zero at two years, the dependence-aware bootstraps do not, and all three agree that the accumulated response excludes zero through the first year. Sign stability across the event and exclusion samples is high, the rolling and recursive windows keep the positive sign in most cases, and the direction of the accumulated path never changes. No other variable combines an early response, a durable accumulation, and this degree of sign coherence.

The contrast with the rest of the credit block is what makes the result specific. Broad NFC and household credit stocks barely move, and their accumulated intervals include zero at every horizon. Easing does not show up as a general expansion of credit. It shows up in the one credit aggregate that finances a collateral-intensive purchase. That is why the housing-credit finding is the center of the thesis rather than a supporting detail inside an asset-price story, and why it is read as a channel-specific statement about transmission and not as a statement about house prices or borrower access, which sit outside the design.

Earlier studies point the same way. Bernanke and Gertler (1995) report, for the United States, that residential investment is the spending component that responds most sharply to a monetary policy shock, and they read that as the signature of the credit channel. They also caution that credit quantities alone cannot show that a credit channel is at work, because quantities mix supply and demand. Jordà et al. (2015) find in a long annual panel of advanced economies that loose monetary conditions feed booms in real estate lending and house prices, and Jordà et al. (2016) show that mortgage lending has grown into the largest part of bank lending. The present result agrees with all three and adds a piece they do not measure. Their identification rests on recursive VAR innovations in the first case and on the policy trilemma in the 2015 panel study, and their housing measures are spending or credit aggregates. Here the same channel is the clearest response to a high-frequency ECB surprise, traced month by month over two years, in a euro-area sample that runs from 2005 to 2025 and covers the financial crisis, the low-rate and QE years, the pandemic, and the tightening that followed.

7.2 Stocks, Flows, and the Lending Result

The housing-credit result rests on a stock-based growth rate, and the choice matters for how it should be read. House-purchase lending growth measures the growth of the outstanding book. It captures cumulative financing conditions and sustained accommodation over the medium run. Pure-new mortgage loans measure monthly originations. They capture immediate lending behavior and are exposed to a short sample, to refinancing composition, and to month-specific reporting noise.

The two series behave as that distinction predicts. The growth of the stock gives the strong and persistent response. New loans are weaker and less stable. Their sample is short, their sign stability is close to a coin toss, and their peak interval includes zero under HAC standard errors even where the bootstrap band is tighter. There is no contradiction in this. Sustained credit accommodation appears in the growth of the housing-credit stock before it appears cleanly in monthly new flows. The two series are also linked in time. An early move in the growth of the stock is associated with a later move in new flows, and the coefficient on that link is positive and precisely estimated. The flow series therefore stays a robustness surface rather than a replacement for the canonical proxy.

The distinction also fixes which literature the result should be compared with. The housing-credit-cycle evidence in Jordà et al. (2016) is built on outstanding mortgage credit, a stock, so the durable stock-based response is the closer counterpart to that work. The noisier flow series has no clean analogue there and is not asked to carry a comparison.

7.3 Financial Intermediation Mechanism

The intermediation evidence is what makes the housing-credit result economically believable. The chain in Equation (3.1) runs from policy news to lending conditions, then to housing credit, and finally to the real-economy block. The lending-condition step is the one that has to hold for the chain to mean anything, and it does. The mortgage and NFC spreads over the deposit facility rate widen after an easing surprise, and their accumulated responses stay positive through two years. The mortgage spread is the most stable response in the whole set, with a perfect sign-stability score and an accumulated interval that excludes zero from one month on under the HAC and the moving-block intervals.

The direction of that move needs one line of explanation, because it is the one place where the thesis appears to conflict with the credit-cost literature. Gertler and Karadi (2015) find that modest surprise movements in the policy rate produce large movements in credit costs, mainly through term premia and credit spreads, and conclude that the standard interest-rate channel misses part of transmission. Their spreads are private borrowing rates over risk-free rates of matching maturity, so a rise means dearer credit. The spreads here are retail lending rates over the policy rate. They widen after an easing surprise while the lending rates themselves show no precise response, so the margin over the policy rate opens. That points to incomplete pass-through rather than a rise in the cost of credit relative to the risk-free curve, and the two findings are complementary. Both say that the policy rate alone does not describe credit conditions and that the intermediary sits between policy and the borrower. The sustained margin matches the intermediary view in Adrian and Shin (2010) and Borio and Zhu (2012), in which policy works through bank balance sheets and loan pricing before it works through volumes. Bank-level euro-area evidence adds that pass-through is uneven. Altavilla et al. (2020) show that pass-through from policy rates to lending rates differs across banks with their capital ratios and sovereign exposures, and that the non-standard measures compressed lending rates most at the weakest banks.

The credit-quantity step is where a naive story breaks. If easing simply expanded all credit, broad NFC and household stocks would move with it. They do not. The picture is sectorally differentiated transmission, not a generic expansion of credit quantities, and it is the pattern the credit-channel theory in Chapter 3 predicts for a collateral-intensive sector.

What the timing outputs add is more modest than the chain might suggest. The sequential pathways in Table 5.1 fix the upstream variable at an early horizon and the downstream variable at a later one, so their order is imposed rather than discovered. Within that design the quantity-to-quantity links are positive and precise, while the link from an early widening of the mortgage margin to later house-purchase growth is negative and marginal. A wider margin in the first month goes with somewhat weaker housing-credit growth six months later, which is the sign a price-quantity relation should carry. The peak horizons tell the same story from another angle. House-purchase lending growth peaks at six months and the spreads at twelve, so the spreads do not lead housing credit. The evidence for intermediation is therefore co-movement on the same shock, a sustained margin and a sustained quantity response over the same year, rather than an observed sequence from one to the other. Monthly aggregates could not separate one bank's decision from one household's response even if the sequence were visible, so the thesis describes a financial propagation with a price side and a quantity side and stops there.

7.4 Equity Responses and Information Effects

The DAX is the hardest financial result to read. It is negative after an easing surprise, and the negativity deepens from three months to two years. The twelve- and twenty-four-month intervals exclude zero at both the 90 and 95 percent levels, and the accumulated path is durable with a perfectly consistent cumulative direction. A response this stable has to be interpreted rather than set aside.

Forcing it into an asset-appreciation story would be wrong, and the literature explains why. An ECB announcement carries two things at once, the policy accommodation and the central bank's own reading of the economy. Jarociński and Karadi (2020) separate the two by the co-movement of interest rates and stock prices inside the announcement window. A pure policy tightening raises rates and lowers equities, while a positive central-bank information shock raises both, and they show that ignoring the information component biases estimates of what policy does. Nakamura and Steinsson (2018) reach the same conclusion for the Federal Reserve from

the reaction of expected output growth to policy surprises. When the information component of an easing surprise is adverse, expected earnings fall, risk premia rise, and the equity market can drop even as policy loosens.

That is the situation the event screen was built for, and the screen says the situation is common. It classifies 132 of the 208 events in the identification window as contaminated or ambiguous, so a large share of the raw target surprises may carry an information component. A negative equity response to a surprise signed as easing is then what the literature predicts, not a contradiction of it. The DAX marks the limit of the result rather than its foundation. The liquidity stock fits the same reading, negative but imprecise, while the housing-credit and lending-condition blocks hold the mechanism.

7.5 Compensation-Linked Responses

The compensation block is weaker, not absent, and that distinction is what the second hypothesis turns on. The real wage tracker excluding one-offs moves at short and intermediate horizons, and its accumulated response excludes zero at six and twelve months under the HAC and wild intervals. By two years that exclusion is gone under every method. The series loses precision exactly where the housing-credit result keeps it. Employment expectations tell a similar story from the other side. They respond early and by a lot, then reverse to a negative and individually significant value at two years. An early reaction that unwinds is not durable income transmission.

The robustness proxies qualify this. The German industry wage-bill measure gives a more positive and durable signal, but it is a national and sectoral series, and Chapter 4 sets out why it stays a secondary proxy.

The weak response is also what earlier work would lead one to expect. Christiano et al. (1999) summarize the VAR evidence for the United States as follows: after a monetary policy shock real wages move only by a small amount, and the price level adjusts with a substantial delay. The euro-area wage tracker also moves only a little after a high-frequency easing surprise, with a short-lived real dip and no precise long-horizon movement. It is also what the staggered-contract logic behind Hypothesis 2 predicts, since negotiated wages can respond to a surprise only as agreements are renewed (Taylor, 1980). What the thesis adds is the comparison. The same shock produces a durable housing-credit response and a fading compensation response, which turns a familiar observation about wages into a statement about channels.

The compensation result is therefore comparative. The euro-area wage tracker and employment expectations respond less consistently and less persistently than house-purchase lending growth and the lending spreads. The German wage bill does not. Whether income gains were absent across households is a different question, one that needs microdata and a different identification design, and Section 7.9 records it.

7.6 Post-COVID and Regime Interpretation

The regime outputs describe historical dependence and do not replace the full-sample estimates. The reason is mechanical. The thesis is about persistence, and persistence needs long horizons. The COVID window has too few observations for any horizon beyond one month, and the tightening regime is short relative to the two-year horizon. A short window cannot identify a two-year response profile, so the regime evidence is kept descriptive by design. This is also the main limit on what the thesis can say about the post-COVID period itself. The months since 2020 are under a third of the sample, so the full-sample estimates describe how ECB policy news transmits across 2005 to 2025, with the post-COVID cycle as one part of that record rather than a separately estimated window.

Read that way, it still adds something. The tightening regime shows positive housing-credit and spread accumulation with negative wage-pressure accumulation, which lines up with the full-sample ordering. The COVID window shows large moves at the impact and one-month horizons across employment expectations, wage pressure, the DAX, and house-purchase lending growth, which may signal state dependence. The pre-QE and QE windows map the same target surprise into very different observed conditions, with strongly positive accumulated house-purchase growth before QE and negative house-purchase growth and NFC spreads under it. The size, and sometimes the sign, of a response can depend on the regime. The full-sample estimates remain the primary result, and the regime evidence bounds how far it should be generalized.

7.7 Implications for Research and Practice

The findings have moderate but real relevance for practice, and the relevance lies in the specific numbers rather than in the general point that channels differ. Three implications follow. First, the horizon at which transmission is assessed changes the verdict. The housing-credit response peaks

at six months, its accumulated value peaks at twelve, and about a quarter of that accumulation reverses by two years. An assessment of credit transmission that stops at one year records the effect at its maximum and overstates its durability, while one that looks only at two years misses the peak. Second, an easing surprise is not passed through one for one to new mortgage borrowers within the window studied. The mortgage margin over the deposit facility rate widens by about 0.05 percentage points one year after a typical surprise, and the two-year estimate is positive but imprecise. Third, the response is sectoral. Broad NFC and household credit stocks do not move with any precision after the same surprise that moves house-purchase lending growth, so a monitoring framework built on total credit would have recorded little, while the channel that responded most durably sat inside it. These implications are analytical rather than prescriptive.

The results also bear on the affordability debate, in a specific and limited way. Chapter 1 set out the backdrop: real German house prices rose by about half between 2015 and 2021 while real compensation barely moved. Any account of that gap that runs through monetary policy has to pass through housing credit, and this thesis studies that step. Easing-signed surprises moved house-purchase lending growth at short and medium horizons and lending margins durably, while they reached the euro-area wage tracker weakly and briefly. That is the credit side of the mechanism the affordability debate presumes, established at monthly frequency with a high-frequency shock. It is not evidence on who obtained credit or on what happened to prices, and it says nothing about access being widened or narrowed for any group. Closing that loop would need three further pieces of evidence: house prices at a frequency that allows a monthly response, loan-level data on who borrowed and at what terms, and household balance-sheet data on who owned housing before the shocks arrived. Without them the thesis stops at the channel, and it says so.

For research, the results carry two lessons. A study that uses raw ECB target surprises as policy shocks should screen for information effects before it reads equity or expectation responses. In this sample 132 of the 208 announcements in the identification window carry a sign conflict or a possible information component, and the raw surprise produces a negative equity response that the screen helps to interpret. And work on pass-through at the monthly frequency gains from treating the margin over the policy rate as the measured object, not only the lending rate. The lending-rate levels in this thesis are estimated imprecisely, while the margin built from them is the most stable response in the whole set, with the same sign in all 42 sample-by-horizon cells, because netting out the policy floor removes the common movement that the rate levels share.

7.8 Limitations

The limits of the design set the limits of the claims. The shock is a high-frequency ECB policy-news surprise, not a purified structural monetary innovation. The best low-frequency bridge, the end-of-month change in the deposit facility rate, has real full-sample relevance but concentrated, event-driven strength, which Chapter 6 documents in detail. That is why the shock is read as policy news rather than as an instrument for a single policy variable, and why information effects and event contamination stay inside the identification environment rather than being assumed away. It also separates this thesis from Gertler and Karadi (2015) and Jarociński and Karadi (2020), who use the surprises inside a VAR to recover structural shocks. The reduced-form route taken here is more transparent about what the surprise is, at the cost of not naming a treatment variable.

The aggregation boundary matters just as much. Quarterly house prices, compensation per employee, and the Bank Lending Survey are not interpolated into monthly responses. Household-level access, debt-service ratios, borrower heterogeneity, regional house prices, tenure status, and bank-level balance sheets are outside the empirical design, and so is any observed sequence from lending conditions to housing credit, as the previous sections made clear. The statistical caveats are the familiar ones. Many response variables, mixed signs, proxy dependence, and horizon dependence all call for caution in reading any single coefficient. The accumulated intervals depend, in addition, on how the dependence across horizons is handled. The HAC intervals treat the horizon estimates as independent, and the joint bootstraps that relax this widen the two-year housing-credit band enough to include zero. The thesis reports both and rests the persistence claim on the horizons where they agree.

Inside those boundaries the conclusion is still substantive. Under a reproducible monthly local-projection framework, ECB policy-news shocks show clearer evidence in house-purchase lending growth and lending spreads than in the euro-area wage tracker and broad productive-credit quantities, with the German industry wage bill as the exception on the compensation side. That is a disciplined result about heterogeneous macro-financial transmission. It is not a welfare or inequality claim.

7.9 Future Research Agenda

Each limitation points to a specific extension. The aggregation boundary comes first. Bank-level and loan-level data would show whether the co-movement of margins and housing credit documented here reflects lender heterogeneity, funding-cost pass-through, margin adjustment, or credit-supply behavior, and would make an actual sequence from lending conditions to credit observable where monthly aggregates cannot. The household boundary comes second. Household-level credit, income, and balance-sheet data would allow incidence analysis across borrowers, renters, homeowners, income groups, and refinancing status, and would turn the comparative compensation result into a statement about who gained. The frequency boundary on house prices comes third. Regional housing data would show whether the housing-credit response is stronger where local housing constraints are already tight, and would connect the credit result to prices without interpolating quarterly series. The identification limits come last. A longer post-COVID sample, and a shock series purged of information effects along the lines of Jarociński and Karadi (2020), would improve the regime analysis and allow the first-stage bridge to the policy rate to be tested again on a sample less dominated by a few events. Each direction relaxes one boundary of the present design without discarding its identification discipline.

8. Conclusion

This thesis asked whether post-COVID ECB monetary easing transmitted more strongly through housing-credit and financial-intermediation channels than through compensation-linked real-economy variables. Within the monthly reduced-form local-projection framework, the answer is a qualified yes. Easing-signed ECB policy news moves house-purchase lending growth clearly at short and medium horizons, and lending margins move with it. Whether this is stronger than transmission to compensation depends on the measure: the euro-area wage tracker responds weakly and briefly, while the German industry wage bill responds durably. The estimates pool easing and tightening surprises across 2005 to 2025, so they do not isolate the post-COVID cycle, which motivates the question rather than defining the sample. The finding concerns where policy news shows up most clearly in aggregate data. It does not show that easing benefits financial markets more than the real economy, and it is narrower than a broad asset-market result.

8.1 Summary of Findings

The full interpretation is developed in Chapter 7. This section restates the main responses in compact form. House-purchase lending growth is the clearest response at short and medium horizons: it reacts on impact, peaks at h6, accumulates through h12, and gives back about a quarter of that accumulation by h24 while remaining positive, with an exclusion of zero that every interval method confirms through h12, which makes it the core empirical contribution. Lending spreads corroborate this reading, as mortgage and NFC spreads are positive and cumulatively durable while broad credit quantities and ECB assets are weaker. The real DAX response is negative at medium and long horizons. It is read as information-effect sensitivity rather than a simple asset-price channel, and therefore bounds broad asset-price claims without displacing the housing-credit evidence. The euro-area compensation proxies respond less consistently. The ECB Wage Tracker excluding one-offs moves at short and intermediate horizons but loses h24 cumulative exclusion. The German industry wage-bill proxy responds positively and durably, though it is a national and sectoral series. The comparison with compensation therefore depends

on the measure: housing credit responds more clearly than the euro-area wage tracker, but not more clearly than the German industry wage bill.

8.2 Interpretation Boundary

These results should be read as reduced-form evidence of heterogeneous monetary transmission. They support a clear short- and medium-horizon response of house-purchase lending growth and a sustained lending-margin response alongside it. They do not support household welfare conclusions, inequality decomposition, redistribution claims, a definitive verdict on household income incidence, or the claim that monetary easing benefits financial markets more than the real economy. The responses are measured in different units and on different samples, so they show where policy news is visible, not who gains from it. This boundary is a source of the design's credibility. The model uses easing-signed high-frequency ECB surprises and dynamic impulse responses to show where policy news stays most visible, and it asks the estimates to identify no more than the design permits.

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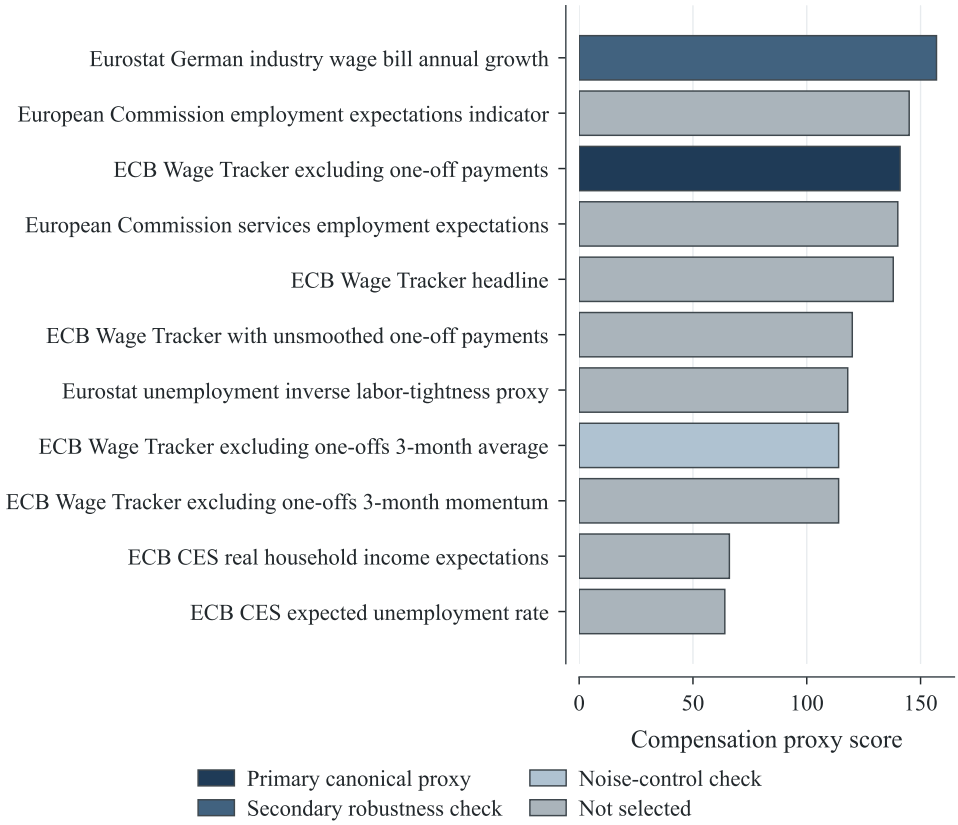
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A. Additional Material

A.1 Additional Figures

The appendix figures document diagnostic and registry outputs that support the chapter-level claims. They do not introduce additional estimands.

Figure A.1: Proxy-validation ranking for compensation-pressure candidates.



Source: own calculations.

A.2 Additional Tables

The table below reports supporting estimates. It adds detail to the response paths discussed in the main text and does not introduce additional estimands.

Table A.1 records the largest response for each variable and the horizon at which it occurs.

Table A.1: Peak response magnitude and timing

Response	Peak h	Peak	90% interval		p	N
house-purchase lending growth	6	0.149	[0.077,	0.221]	0.001	240
pure new house-purchase loans	24	−0.031	[−0.075,	0.014]	0.253	73
mortgage lending spread	12	0.050	[0.007,	0.093]	0.054	234
NFC lending spread	12	0.025	[0.006,	0.045]	0.034	234
NFC credit	24	−0.001	[−0.004,	0.002]	0.561	222
real DAX	24	−0.027	[−0.049,	−0.006]	0.037	222
ECB assets	24	−0.011	[−0.035,	0.014]	0.476	222
real wage tracker excl. one-offs	24	0.260	[−0.099,	0.620]	0.233	128
German industry real wage bill growth	1	0.422	[−0.023,	0.867]	0.119	234
employment expectations	1	1.173	[−0.505,	2.851]	0.250	245

Notes: Largest absolute normalized response across the reported horizons, with its HAC 90 percent interval and estimation sample size. Source: own calculations.

A.3 Technical Derivations

This appendix records notation conventions used in the empirical chapters. The derivations support response-path reporting and do not add structural identifying assumptions.

A.3.1 Cumulative Responses

For response i, the cumulative response through reported horizon H is:

$$\mathrm{CIRF}_i(H) = \sum_{h \in \mathscr{H}: h \le H} \widehat{\mathrm{IRF}}_i(h), \tag{A.1}$$

where $\mathscr{H} = \{0, 1, 3, 6, 12, 24\}$ is the thesis-facing horizon set. This convention matches the cumulative-response summaries exported by the empirical pipeline and used in the persistence tables.

The HAC interval around $\mathrm{CIRF}_i(H)$ uses the standard error in Equation (3.8), which treats the horizon-level estimates as independent. The dependence-aware alternative resamples the residuals of all horizons jointly, in moving blocks of twelve months or with wild sign flips, re-estimates every horizon on the same draw, and takes percentiles of the resulting running sums. Table 6.4 reports those intervals next to the HAC intervals of Table 5.4.

A.4 Data Sources

Table A.2 lists every series used in the estimation with its provider, series code, and access date. Two ECB series and the German property-price index were downloaded through FRED, which redistributes them unchanged.

Table A.2: Data sources, series codes, and access dates

Variable	Source and series code	Accessed
Policy shock and announcement surprises		
Target, timing, FG, and QE factors	Ragusa (n.d.), factor vintage of 30 Oct 2025, built with the method of Altavilla et al. (2019)[a]	15 May 2026
Announcement-window OIS and STOXX 50 surprises	European Central Bank (n.d.-d)[a]	15 May 2026
Policy rate and liquidity		
Deposit facility rate, daily	ECB, `FM.D.U2.EUR.4F.KR.DFR.LEV`, retrieved as `ECBDFR` (European Central Bank, n.d.-c)[a]	15 May 2026
Eurosystem total assets, weekly	ECB, `ILM.W.U2.C.T000000.Z5.Z01`, retrieved as `ECBASSETSW` (European Central Bank, n.d.-a)[a]	15 May 2026
Wu–Xia shadow rate	Wu and Xia (n.d.), method of Wu and Xia (2020)[a]	15 May 2026
Lending conditions and credit (European Central Bank, n.d.-b)		
Mortgage lending rate	`MIR.M.U2.B.A2C.A.R.A.2250.EUR.N`[a]	16 May 2026
NFC lending rate	`MIR.M.U2.B.A2A.A.R.A.2240.EUR.N`[b]	16 May 2026
House-purchase lending growth	`BSI.M.U2.Y.U.A22.A.I.U2.2250.Z01.A`[a]	16 May 2026
Pure new house-purchase loans	`MIR.M.U2.B.A2C.A.B.A.2250.EUR.P`[a]	16 May 2026
Household loan stock	`BSI.M.U2.N.A.A20T.A.1.U2.2250.Z01.E`[b]	15 May 2026
NFC loan stock	`BSI.M.U2.N.A.A20T.A.1.U2.2240.Z01.E`[b]	15 May 2026
Bank Lending Survey standards and demand	`BLS.Q.U2.ALL.{Z.H.H, O.E.Z, Z.H.C}.B3.{ST.S, ZZ.D}.WFNET`[a]	16 May 2026
Asset prices and deflators		
DAX close, monthly	Yahoo Finance (n.d.), `^GDAXI`[a]	15 May 2026
German residential property prices	BIS, `Q:DE:N:628`, retrieved as `QDEN628BIS` (Bank for International Settlements, n.d.)[b]	15 May 2026
HICP, Germany, EA20, and euro area	Eurostat (n.d.-b), index 2005 = 100, all items[a]	15 May 2026
Compensation and labour market		
ECB wage tracker (ex one-offs, headline, unsmoothed)	`EWT.M.U2.N.WT.{INWX, INWS, INWR}._T.4F0.GY` (European Central Bank, n.d.-b)[b]	16 May 2026
Compensation per employee, EA20	`MNA.Q.Y.I9.W2.S1.S1._Z.COM_PS._Z._T._Z.IX.V.N` (European Central Bank, n.d.-b)[b]	15 May 2026
Consumer income and unemployment expectations	`CES.M.Z18.ALL.T.{C3220, C4031}.NUM_VAR.WM` (European Central Bank, n.d.-b)[a]	16 May 2026
German industry wage bill	`sts_inlb_m`, DE, B–E36, NSA (Eurostat, n.d.-a)[a]	16 May 2026
German retail trade volume	`sts_trtu_m`, DE, volume of sales (Eurostat, n.d.-a)	15 May 2026
Employment expectations, EA20	`ei_bsee_m_r2`, EEI and services (Eurostat, n.d.-a)[a]	16 May 2026
Unemployment rate, EA21	`une_rt_m`, seasonally adjusted (Eurostat, n.d.-a)[b]	16 May 2026

Notes: [a] Identical to the live source, at the precision stored in the raw file, when re-checked on 27 September 2026. [b] Same series, revised by the provider after the access date. The thesis uses the vintage downloaded on the access date. Source: own compilation.

A.5 Canonical Artifact Registry

Every estimate in this thesis is generated by the pipeline in scripts/run_full_pipeline.py and written to results/final/. The data-source, software, registry, persistence-synthesis and hypothesis-assessment tables and the mechanism diagram are written by hand from those outputs. The registry below lists the canonical output files behind the reported evidence so that each result can be traced back to the artifact that produced it.

Table A.3: Canonical pipeline artifacts underlying the reported results

Artifact	Path
Canonical monthly model dataset	data/processed/eu_de/final_monthly_model_dataset.csv
Monthly variable dictionary	data/processed/eu_de/monthly_variable_dictionary.csv
Information-effect event screen	results/final/diagnostics/information_effect_event_screen.csv
Instrument strength matrix	results/final/diagnostics/instrument_strength_matrix.csv
Monthly first-stage tournament	results/final/diagnostics/monthly_first_stage_tournament.csv
Banking timing matrix	results/final/mechanism/banking_timing_matrix.csv
Sequential transmission outputs	results/final/mechanism/sequential_timing_outputs.csv
Regime-specific reduced-form LP output	results/final/regime/monthly_regime_reduced_form_lp.csv
OLS policy-rate direction check	results/final/robustness/monthly_ols_policy_rate_comparison.csv
Cumulative stability outputs	results/final/stability/monthly_cumulative_stability_long.csv
Directional stability matrix	results/final/stability/monthly_directional_stability_matrix.csv
Rolling and recursive LP stability	results/final/stability/monthly_stability_metrics.csv
Cumulative persistence table	results/final/tables/cumulative_persistence_table.csv
Horizon significance table	results/final/tables/horizon_significance_table.csv
Normalized baseline IRF output	results/final/tables/normalized_irf_outputs.csv
Peak response table	results/final/tables/peak_response_table.csv
Stability metrics	results/final/tables/stability_metrics.csv
Bootstrap sensitivity summary	results/final/uncertainty/bootstrap_sensitivity_summary.csv
Horizon dependence diagnostics	results/final/uncertainty/horizon_dependence_diagnostics.csv

Source: own compilation.

A.6 Software and Tools

Table A.4 lists the software used to build the datasets, estimate the results, and typeset the thesis, and the data portals the raw data come from.

Table A.4: Software, tools and data portals used in the preparation of this thesis

Tool	Version	Use
Python	3.13.11	Shock construction, local projections, bootstraps, diagnostics, thesis tables and figures
numpy, pandas, scipy	2.4.1, 3.0.0, 1.17.0	Numerical work and data handling
statsmodels	0.14.6	Regression estimation with HAC covariance
matplotlib	3.10.8	Figures
R	4.4.2	Construction of the processed datasets, two supplementary figures
readxl, dplyr, tidyr	1.4.3, 1.1.4, 1.3.1	Data import and transformation
jsonlite, ggplot2	1.8.9, 3.5.1	Configuration files and plotting
XeLaTeX (TeX Live 2026), biblatex with biber	biber 2.21	Typesetting and reference management
Data portals		
ECB Data Portal (data.ecb.europa.eu)		
Eurostat Database (ec.europa.eu/eurostat)		
FRED, Federal Reserve Bank of St. Louis (fred.stlouisfed.org)		
ECB Euro Area Monetary Policy Event-Study Database (ecb.europa.eu)		
Monetary policy factors (gragusa.org/factors)		
Wu-Xia shadow rate (sites.google.com/view/jingcynthiawu)		
Yahoo Finance (finance.yahoo.com)		

Source: own compilation.

Statement on the Use of AI Tools

I used AI tools in preparing this thesis, specifically Claude (Anthropic) and, for retrieving the raw data, Codex (OpenAI). I used them for four purposes:

1. **Code.** Writing, debugging and documenting the Python and R code for data construction, estimation, figures and tables, and the LaTeX source and build setup.
2. **Building the analysis.** Implementing the empirical steps I specified, and checking that results, tables and text agree with each other.
3. **Organising information.** Structuring my notes and literature, and checking references and data provenance against the original sources.
4. **LaTeX design and text.** Designing the LaTeX layout, and suggesting wording for parts of the text, which I then revised.

The research question, the empirical design, the choice of data, variables and specifications, the interpretation of the results and the conclusions are my own. I reviewed every result and every sentence against the underlying outputs before including it, and I take full responsibility for the content of this thesis, including any errors that arise from the use of AI tools.

Tübingen, 15/10/2026

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Chirag Khanna

Eidesstattliche Erklärung

Ich versichere an Eides Statt, dass ich die Arbeit selbständig verfasst, keine anderen als die angegebenen Hilfsmittel und Quellen benutzt habe, alle wörtlich oder sinngemäß aus anderen Werken übernommenen Aussagen als solche gekennzeichnet habe und dass die Arbeit weder vollständig noch in wesentlichen Teilen Gegenstand eines anderen Prüfungsverfahrens gewesen ist und dass ich die Arbeit weder vollständig noch in wesentlichen Teilen bereits veröffentlicht habe sowie dass das in Dateiform eingereichte Exemplar mit den eingereichten gebundenen Exemplaren übereinstimmt.

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Chirag Khanna